Forecasting Firm-Level Tail Risk with Structured Mixture Density Networks*

Jianbin Chen†

October 20, 2026

Click here for the latest version

Abstract

This paper develops a structured mixture density network to forecast individual stock return distributions across three possible future market environments: adverse, middle, and favorable. In the structured models, the probabilities of the three market environments depend only on macroeconomic information. Within each environment, each firm's return follows a Normal distribution whose mean and variance depend on firm characteristics. For U.S. stock forecasts formed from 2007 through 2024, the structured models outperform all four benchmarks on every out-of-sample density and tail forecast score. Stocks with lower expected shortfall (ES) predicted by the structured models earn significantly lower realized returns in bad market months. The structured models allow each stock's ES to be decomposed into contributions from the adverse, middle, and favorable market environments. After adjusting the adverse environment's contribution for total ES and other risk measures, stocks in the bottom decile of the adjusted contribution have significantly higher crash rates than those in the top decile in bad market months. The structured models therefore provide more accurate distribution forecasts and additional information about stocks' crash risk in bad market months.

Keywords: expected shortfall, density forecasting, mixture density networks, downside exposure, macroeconomic conditioning.
JEL codes: C45, C53, C58, G11, G12.

*Job market paper.

†University of California, Riverside. Email: jchen939@ucr.edu.

1 Introduction

Firms face common credit conditions, liquidity, and aggregate activity, but their returns can respond differently to the same conditions. Differences in balance sheets, profitability, and valuation can produce different return outcomes within a common economic environment. Research on common tail risk and downside exposure highlights both the shared environment and firms' different responses (Kelly and Jiang, 2014; Ang et al., 2006a). A model of next month's stock returns therefore needs to describe both uncertainty about the common environment and differences across firms within that environment. I study whether giving macroeconomic and firm information separate roles in a neural forecasting model improves predictions of individual stock return distributions.

I develop a structured mixture density network (MDN). An MDN uses observed predictors to forecast a probability distribution for next month's return (Bishop, 1994). My model represents the future environment with three possible states. Macroeconomic information alone determines their probabilities, and all firms face the same state realization. Within each state, each firm's return follows its own Normal distribution. Firm characteristics predict its mean and standard deviation. Thus, the same environment can imply different return distributions for different firms. Returns are assumed independent across firms conditional on the state and observed predictors. Before the state is known, uncertainty about the shared environment can make returns dependent across firms.

The common state also determines how the model is trained. For each month, the joint likelihood first evaluates all firms' observed returns together under each possible state, then averages the three state-specific likelihoods using the predicted state probabilities. This makes the assumption of a shared state part of estimation. The training objective combines a normalized joint negative log likelihood with a marginal negative log likelihood that evaluates individual stock forecasts.

I evaluate three structured specifications. Structured K3 predicts component means and standard deviations from firm information. FiLM Sigma uses feature-wise linear modulation (FiLM) to let macroeconomic information also affect the standard deviations, while means depend only on firm information. FiLM All allows both firm and macroeconomic information to affect means and standard deviations. All three retain the same common-state structure and macro-driven state probabilities. These specifications allow me to examine whether the forecasting advantage appears across different ways of using macroeconomic information within states.

The out-of-sample comparison covers U.S. stock forecasts formed from January 2007 through December 2024, giving 216 evaluation months. I compare the three structured

specifications with four benchmarks: Rolling Normal, Linear Quantile, Fused K1, and Fused K3. The fused neural benchmarks combine macroeconomic and firm inputs in one network and use one or three Normal components. The three structured models occupy the top three positions on every forecast score. Each has significantly lower forecast losses than both fused benchmarks on all eight scores, covering the full distribution and its lower tail. The forecasting gains therefore occur across all three structured specifications and across several measures of forecast accuracy.

I then use the predicted distributions to study subsequent stock losses. I first measure tail risk using expected shortfall (ES), the average return in the worst 5% or 10% of a stock's predicted return distribution. Higher ES indicates a better predicted outcome in this tail. Each month, I rank stocks by ES using information available at month-end and form ten portfolios for the following month. I evaluate their realized returns separately in bad market months and other months. Bad months are the 20% of evaluation months with the lowest realized market excess returns. This classification is made after the stock portfolios have been formed.

Total ES identifies stocks that perform worse when the market performs poorly. In bad market months, the highest-ES decile outperforms the lowest-ES decile by about 6% per month under equal weighting. The result holds for all three structured specifications and both ES levels, and it also holds with value weights. In other months, the average return differences are not significant. Separate monthly regressions associate lower ES with higher crash probabilities and larger losses beyond a 10% return decline in both bad and other months, after controlling for historical stock characteristics and industry. A crash is a next-month stock return below -10%. Together, these results show that total ES forecasts help identify stocks that subsequently suffer larger losses.

The structured model also allows me to compare how much of each stock's ES comes from the same market environment. I label the three components adverse, middle, and favorable using their average predicted means before each test year. Each stock's ES is the sum of the three component contributions. I call the contribution from the adverse state the *adverse contribution*. Two stocks can have similar total ES even when the adverse state contributes different amounts to their predicted tail losses. I test whether this difference helps explain subsequent stock losses after accounting for total ES and other risk measures.

Monthly cross-sectional regressions show that adverse contribution provides additional information. After controlling for total ES, industry, historical return characteristics, market and downside exposures, and predicted mean and volatility, lower adverse contribution is more strongly associated with higher crash probabilities in bad market months than in other months. The corresponding relations with lower returns and larger losses

beyond the 10% threshold are also stronger in bad months. These tests show that the adverse contribution contains information beyond its linear relations with total ES and the included controls.

Each month, I regress adverse contribution on total ES and the controls, then subtract the fitted value from each stock's adverse contribution to obtain a residual. A lower residual means that the stock's adverse contribution is more negative than the regression predicts from its total ES and other characteristics. Before observing next-month returns, I select the bottom and top 10% of stocks ranked by these residuals.

In bad market months, the bottom group has a crash rate about 6–8 percentage points higher than the top group across all three structured specifications and both ES levels. No significant average gap is found in other months. In bad market months, the two groups have similar average total ES, and the bottom group has slightly lower average market beta. The residual ranking therefore identifies stocks with higher crash rates even though they do not consistently have worse average ES or higher market beta.

The main findings persist under additional training and inference checks. All three structured models retain their forecast advantage over the fused benchmarks across random seeds and under fixed-duration training. The higher bad-month crash rates in the bottom residual group also persist. The contribution-regression results remain significant under alternative definitions of bad market months and different Newey–West lag lengths.

The paper makes two contributions. First, it develops and evaluates a structured MDN framework that combines macro-driven common state probabilities, firm-specific conditional return distributions, and a common-state training objective. The evidence supports this framework as an effective way to use macroeconomic and firm information for return-distribution forecasting. Second, the tail-risk application shows that the forecasts provide information about subsequent stock losses. Total ES identifies stocks with worse downside performance, and the adverse-state contribution identifies further differences in crash risk after controlling linearly for total ES and other risk measures. The common states make these contributions comparable across firms, linking the model's structure to the economic information in its forecasts.

1.1 Related Literature

This paper draws on research on return-distribution forecasting, common risks, and firm-level downside risk. The structured MDN separates common conditions from firm-specific outcomes. Its forecasts allow me to test the information in both total tail risk and its component contributions.

1.1.1 Machine Learning and Return-Distribution Forecasts

Gu et al. (2020) show that machine learning can improve expected-return forecasts by capturing nonlinear predictor interactions. A related line of research incorporates financial structure into statistical learning. Kelly et al. (2019) use firm characteristics to model changing exposures to common latent factors. Gu et al. (2021) extend this approach with nonlinear characteristic-dependent exposures in an autoencoder asset-pricing model. These studies show how common sources of return variation can be combined with differences across firms within a structured model.

Distribution forecasts describe the range of possible returns and their probabilities. Closely related to this paper, Baruník et al. (2026) use quantile neural networks and spline interpolation to forecast stock return distributions from firm and market predictors. Their approach provides forecasts of quantiles and distributional moments and captures non-Gaussian features such as asymmetry. I study how an explicit division between common and firm-level information improves density forecasts. Macroeconomic inputs determine a single set of monthly component probabilities. A firm network predicts each stock's component means and standard deviations; in the FiLM specifications, macroeconomic inputs also adjust the learned firm features used for these predictions. This produces a smooth predictive density whose tail loss can be separated into component contributions.

1.1.2 Mixture Models and Shared States

Conditional-density models describe dynamics beyond the mean and variance; Hansen (1994) develops an autoregressive framework for this purpose. The neural mixture method builds on Bishop (1994), who combines neural networks with mixture densities to model conditional distributions. The related mixture-of-experts literature makes both mixing weights and component predictions depend on inputs (Jordan and Jacobs, 1994). In financial econometrics, Giacomini et al. (2008) study flexible mixtures of t-distributions and demonstrate their usefulness in density forecasting and option pricing. Mixtures provide a tractable way to represent asymmetry, multiple peaks, and other departures from a Normal distribution.

Common latent states also have a long history in economic and financial modeling. Hamilton (1989) models discrete changes in an economic time series through an unobserved Markov state. Gray (1996) uses regime switching to model the conditional distribution of interest rates, and Ang and Bekaert (2002) study international asset allocation with regime-dependent return distributions. These models motivate describing changes in the economic environment through a small number of common states.

The structured MDN applies one common state to each month's stock returns. Macroeconomic inputs determine the state probabilities, and firm characteristics determine each firm's return distribution within each state. The joint training objective evaluates all firms together under each possible state. Averaging over the uncertain common state can then induce dependence across firms. The FiLM specifications use the feature adjustments of Perez et al. (2018) to let macroeconomic conditions also affect the learned firm features used to predict component parameters. All three specifications retain the same economic structure.

1.1.3 Tail Risk, Distribution Shape, and Stock Losses

Kelly and Jiang (2014) use firm-level crashes to estimate common variation in tail risk and study its relation to asset prices. Ang et al. (2006a) examine the relation between expected returns and sensitivity to downside market movements. Ang and Chen (2002) also document stronger stock–market correlations during downside moves. These studies motivate a common description of market conditions together with measures of how stocks differ in their exposure to adverse outcomes. My model implements this division within a predictive density: state probabilities are shared, and the return distribution within each state varies across firms.

The economic relevance of distribution shape also appears in research on higher moments. Harvey and Siddique (2000) study conditional skewness in asset-pricing tests, while Dittmar (2002) examines nonlinear pricing kernels and preferences involving higher moments. These studies motivate looking beyond mean and variance when describing stock risk. My moment-matched comparisons address a forecasting question: whether the fitted mixture improves predictions when its mean and variance are held fixed. The contribution tests then examine the economic information in the composition of its predicted tail losses.

Firm-level loss predictability has also been studied through distributional asymmetry. Chen et al. (2001) relate future return skewness to past returns and trading volume. Tail-risk measures contain information about subsequent returns as well. Huang et al. (2012) document a positive expected-return premium for their extreme downside-risk measure, while Atilgan et al. (2020) study left-tail momentum and its connection to underreaction and costly arbitrage. My analysis starts with forecasts of next-month return distributions and examines the stock losses that follow. Total-ES portfolios test whether the rankings distinguish downside performance. The contribution regressions and residual groups then ask whether tail composition adds information after accounting for total ES, historical market and downside exposures, and predicted mean and volatility. This connects a forecast's component structure to realized vulnerability in bad market months.

1.1.4 Expected Shortfall, Risk Attribution, and Forecast Evaluation

The coherent-risk literature provides a foundation for measuring tail losses with ES (Artzner et al., 1999; Acerbi and Tasche, 2002). Rockafellar and Uryasev (2000) develop conditional value-at-risk for portfolio risk management and optimization. Risk-allocation research studies how an aggregate risk measure can be divided among its sources; Denault (2001) examines allocation across portfolios. Here I divide a stock's ES among the components of its predictive mixture. Each contribution combines the component's share of lower-tail outcomes with its average return in that tail. The empirical question is whether these contributions help explain subsequent stock losses after accounting for total ES.

A substantial literature forecasts conditional tail measures directly. Engle and Manganelli (2004) model the dynamics of conditional quantiles using regression quantiles. McNeil and Frey (2000) combine conditional volatility modeling with extreme-value methods to estimate conditional VaR and ES. Taylor (2019) and Patton et al. (2019) develop approaches to joint VaR–ES modeling and evaluation. My structured MDN obtains these tail measures from a forecast of the full density, which also supplies component-level tail weights and losses.

The evaluation draws on several complementary approaches. Proper scoring rules assess the full distribution (Gneiting and Raftery, 2007). Gneiting and Ranjan (2011) develop threshold- and quantile-weighted scores, while Diks et al. (2011) propose conditional- and censored-likelihood scores for selected regions of a density. Calibration diagnostics assess how forecast probabilities agree with realized outcomes (Diebold et al., 1998; Gneiting et al., 2007). Fissler and Ziegel (2016) establish scoring rules for evaluating VaR and ES together. Predictive-ability testing provides a framework for comparing forecasting procedures out of sample (Diebold and Mariano, 1995; Giacomini and White, 2006). I combine density and tail-score comparisons with the stock-loss tests to assess both forecast accuracy and the economic information in the ES decomposition.

2 Data and Methods

2.1 Data

The estimation sample covers formation months from January 2000 through December 2024. Monthly returns, prices, market capitalization, and trading volume come from the Center for Research in Security Prices (CRSP). Accounting ratios and industry information come from Compustat through Wharton Research Data Services (WRDS). An observation must have a firm identifier, a valid formation-month return, positive capitalization, and a

price of at least \$5. I exclude financial firms with Standard Industrial Classification (SIC) codes 6000–6999. The models use 21 continuous firm predictors: 15 accounting ratios, five derived market variables, and the raw stock return in the formation month. For the neural models, industry enters through a six-dimensional learned embedding based on the Fama–French 12-industry classification. Missing industry classifications have a separate category.

Table 1: Firm-level information used in the density models

Group	Predictors
Valuation	Book-to-market, enterprise-value multiple, price-to-sales, price-to-cash-flow
Profitability and efficiency	Operating margin, cash-flow margin, return on equity, gross profitability, asset turnover
Leverage and liquidity	Debt-to-assets, short-term debt share, cash ratio, current ratio
Investment and accounting	R&D-to-sales, accruals
Market characteristics	Current monthly return, skip-month momentum, historical mean, historical volatility, turnover, log capitalization

Notes: The first four groups contain 15 accounting ratios. The last contains five derived market variables and the raw formation-month return, giving 21 continuous firm inputs. Industry enters separately through six learned features. Price is used for construction and eligibility, not as a separate neural input. Appendix A lists the fields.

I calculate historical variables on a complete calendar-month grid for each stock before applying the sample filters. A missing month therefore remains missing instead of being treated as the next observed record. The target is the raw CRSP return in the next calendar month, using the `mthret` field in the WRDS CRSP version-2 monthly data. I check that stock–month records are unique and that the evaluation returns match the corresponding CRSP calendar months. The current return is observed at formation and enters directly as a predictor; it is distinct from this next-month target. The formation-month return and the momentum predictor capture return history at different horizons, motivated by short-horizon predictability and intermediate-horizon momentum (Jegadeesh, 1990; Jegadeesh and Titman, 1993). Momentum compounds returns over the preceding 11 months, excluding the formation month, and requires at least six valid observations. Historical mean and volatility use the 12 months ending at formation and also require at least six observations. Turnover is monthly volume divided by shares outstanding in consistent units. Size is log market capitalization.

Missing accounting ratios are filled first with the median for the same industry and

formation month. If that median is unavailable, I use the median for all eligible firms in that month, then zero if needed. The ratios are winsorized: values below the monthly 1st percentile are set to that cutoff, and values above the 99th percentile are set to the upper cutoff. Missing derived variables use formation-month market medians; any remaining missing inputs use training-window medians. All continuous inputs are scaled with parameters estimated from the training sample. Appendix A lists the variables and explains their construction.

The macroeconomic inputs are 13 variables covering real activity, inflation, interest rates, credit markets, and equity markets. The monthly macroeconomic database FRED-MD, based on Federal Reserve Economic Data (FRED), supplies monthly vintage files; its variables use the specified transformations (McCracken and Ng, 2016). For the early years, I use historical vintages reconstructed from archived data available by the corresponding month-end. Each forecast month uses the latest nonmissing transformed observation in its corresponding vintage, preserving the different reporting lags across series. Additional financial series come from FRED, and the S&P 500 return comes from WRDS. Accounting data are matched using their recorded public dates. The market excess-return series from the Fama–French data is used to classify realized market conditions and estimate historical market and downside exposures.

Table 2: Forecast and empirical evaluation samples

Sample	Stock–months	Months
Common seven-model forecast sample	480,446	216
Common eligible ES portfolio sample	480,429	216
Common regression sample with valid historical controls	410,049	216
Eligible rows excluded by historical requirements	70,380	—

Notes: Dates refer to formation months. Returns are observed from February 2007 through January 2025. Portfolio tests add date-matched CRSP ordinary-share and exchange requirements. Regressions also require valid historical market beta, downside beta, idiosyncratic volatility, and all controls. Missing betas are not replaced with zero.

All seven models are compared on the same 480,446 stock–months over 216 formation months, from January 2007 through December 2024. The corresponding realized returns run from February 2007 through January 2025. Portfolio tests also require date-matched CRSP ordinary shares listed on the New York Stock Exchange (NYSE), the American Stock Exchange (AMEX), or Nasdaq. This leaves 480,429 observations. The conditional regressions require valid historical exposures and a complete control set, leaving 410,049 stock–months. All three structured specifications and every control specification use this same regression sample. The historical requirements remove stocks with insufficient

histories in individual months.

2.2 Mixture Density Forecasts

2.2.1 Normal Mixtures

I use i to index stocks, t for the formation month, and $t+1$ for the next calendar month. The realized raw stock return is $R_{i,t+1}$; returns enter the formulas as decimals, so -0.10 means -10%. The information available at formation is denoted by $\mathcal{F}_t$. All predictive distributions below condition on this information, even when the conditioning argument is omitted.

A Normal mixture is a weighted average of K Normal component distributions, indexed by $k = 1, \ldots, K$. Each component has its own mean and standard deviation, and the weights are nonnegative and sum to one. For a possible return value r, stock i's predictive density is

$$f_{i,t}(r) = \sum_{k=1}^{K} \pi_{i,k,t} \frac{1}{\sigma_{i,k,t}} \phi\left(\frac{r - \mu_{i,k,t}}{\sigma_{i,k,t}}\right), \qquad \pi_{i,k,t} \geq 0, \quad \sum_{k} \pi_{i,k,t} = 1. \tag{1}$$

Here ϕ is the standard Normal density, $\pi_{i,k,t}$ is the weight on component k, $\mu_{i,k,t}$ is its conditional mean, and $\sigma_{i,k,t} > 0$ is its conditional standard deviation, also called its scale. Thus $\sigma^2_{i,k,t}$ is the component variance. A sum over k includes all K components.

The weight $\pi_{i,k,t}$ gives the probability associated with component k. Conditional on this component, the return is Normally distributed with mean $\mu_{i,k,t}$ and standard deviation $\sigma_{i,k,t}$. The predictive density in equation (1) combines these component distributions using their probabilities as weights. When $K = 1$, it reduces to a single Normal distribution. With multiple components, differences in means and scales can produce asymmetry or more than one peak. Although each component is Normal, the mixture need not be.

The predictive cumulative distribution function (CDF) gives the probability of a return at or below r:

$$F_{i,t}(r) = \Pr(R_{i,t+1} \leq r \mid \mathcal{F}_t) = \sum_{k} \pi_{i,k,t} \Phi\left(\frac{r - \mu_{i,k,t}}{\sigma_{i,k,t}}\right),$$

where Φ is the standard Normal CDF and Pr denotes probability. The quantile function is $Q_{i,t}(u) = F^{-1}_{i,t}(u)$ for a probability level $0 < u < 1$. It returns the cutoff below which the distribution places probability u.

Let $\widehat{m}_{i,t}$ and $\widehat{v}_{i,t}$ denote the predictive mean and standard deviation implied by the mixture. The hats indicate quantities calculated from the estimated component parameters

and weights:

$$\widehat{m}_{i,t} = \sum_k \pi_{i,k,t}\mu_{i,k,t}, \tag{2}$$

$$\widehat{v}_{i,t} = \left[\sum_k \pi_{i,k,t}(\sigma^2_{i,k,t} + \mu^2_{i,k,t}) - \widehat{m}^2_{i,t}\right]^{1/2}. \tag{3}$$

The mixture mean is the weighted average of the component means. The mixture variance equals the weighted average of component variances plus the weighted variance of component means.

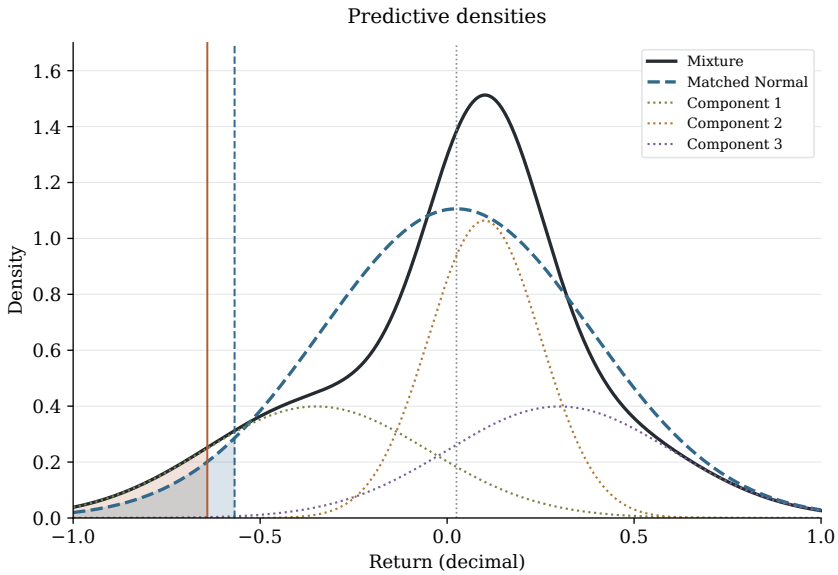
The same distribution also determines lower-tail risk measures. Throughout, VaR and expected shortfall (ES) are expressed as returns rather than positive loss amounts. For a tail probability $0 < \alpha < 1$, the lower-tail quantile, or VaR, is $q_{i,t,\alpha} = Q_{i,t}(\alpha)$. Expected shortfall is the average predicted return below that cutoff:

$$\mathrm{ES}_{i,t,\alpha} = \mathbb{E}_t[R_{i,t+1} \mid R_{i,t+1} \le q_{i,t,\alpha}] = \frac{1}{\alpha}\int_0^\alpha Q_{i,t}(u)\,du. \tag{4}$$

The operator $\mathbb{E}_t[\cdot]$ denotes expectation conditional on $\mathcal{F}_t$; the additional condition restricts returns to the predicted lower tail. For example, at $\alpha = 0.05$, VaR is the fifth percentile and ES averages returns in the lowest 5% of the distribution. A more negative ES means a more severe predicted tail loss.

Figure 1 illustrates why the mean and variance do not fully describe tail risk. The mixture and the single Normal distribution have the same mean and variance, but different lower-tail cutoffs and average tail returns.

Figure 1: Tail Risk in a Normal Mixture and a Moment-Matched Normal



Moments and tail measures

Common moments

Mean	0.025
Standard deviation	0.361

5% lower tail

	Mixture	Normal
VaR	-0.641	-0.568
ES	-0.800	-0.719

Notes: Colored dotted curves show the weighted Normal components; the solid black curve is their mixture. The blue dashed curve is a single Normal with the same mean and standard deviation. The orange solid and blue dashed vertical lines mark the respective 5% VaR cutoffs; the shaded regions show the lower tails. The gray dotted line marks the common mean. The right panel reports the common moments and the different tail measures. Returns are in decimal units. This is an illustrative example.

2.2.2 Structured MDN Parameterization

A Normal mixture describes a return distribution through its component probabilities, means, and standard deviations. A mixture density network (MDN) uses neural networks to predict these parameters from observed information. I propose a structured MDN that organizes this information around a simple economic idea: firms face a common market environment, but their returns can respond differently to it.

For example, tighter credit conditions may affect many firms at once, while differences in leverage, profitability, and liquidity help explain why some firms are more vulnerable than others. The model gives macroeconomic and firm information distinct roles. Macroeconomic variables describe which return environment is more likely next month. Firm characteristics describe the range of returns each firm may experience within that environment. This separates uncertainty about the common environment from differences in firms' outcomes within it. The design uses the same macroeconomic information consistently across the stock panel while retaining differences across firms.

I evaluate three structured specifications: Structured K3, FiLM Sigma, and FiLM All. All three represent the future environment with three possible states, and all firms face the same state realization. Macroeconomic inputs determine the common state probabilities.

In Structured K3, firm inputs determine each firm's mean and standard deviation in each state. FiLM Sigma also uses macroeconomic information to adjust the learned firm features used for component standard deviations. FiLM All uses this adjustment for both component means and standard deviations. All three specifications therefore allow firms to have different return distributions within the same common state.

To express this design formally, let $x_{i,t} \in \mathbb{R}^{21}$ be the vector of 21 cleaned and scaled firm predictors, $e_{i,t} \in \mathbb{R}^6$ the six-dimensional learned industry embedding, and $m_t \in \mathbb{R}^{13}$ the vector of scaled macroeconomic predictors. The industry embedding represents each industry with six learned numerical features. The firm network receives the 21 firm predictors together with these six industry features, giving 27 inputs. The macro network receives the 13 macroeconomic predictors. Figure 2 shows these two paths. In Structured K3, they determine the mixture parameters as follows:

$$\pi_{k,t} = \pi_k(m_t), \qquad \mu_{i,k,t} = \mu_k(x_{i,t}, e_{i,t}), \qquad \sigma_{i,k,t} = \sigma_k(x_{i,t}, e_{i,t}). \tag{5}$$

The functions π_k, μ_k, and σ_k are learned by the networks. The probability $\pi_{k,t}$ has no firm index because every firm uses the same probability for state k in month t. The means $\mu_{i,k,t}$ and standard deviations $\sigma_{i,k,t}$ carry a firm index because they describe that firm's return within the state. Each firm's predictive density is

$$f_{i,t}(r \mid \mathcal{F}_t) = \sum_{k=1}^{3} \pi_{k,t} \mathcal{N}(r; \mu_{i,k,t}, \sigma^2_{i,k,t}). \tag{6}$$

Here $\mathcal{N}(r; \mu, \sigma^2)$ is a Normal density evaluated at r, with mean μ and variance σ^2. A softmax transformation converts the macro network's three output scores into nonnegative probabilities that sum to one. The mixture averages the firm's three possible return distributions using these common probabilities.

The three mixture components correspond to three possible realizations of a common future state, $S_{t+1} \in \{1, 2, 3\}$, with probabilities $\pi_{k,t}$. Here state refers to the common environment, while component refers to the corresponding Normal distribution in a stock's mixture. Conditional on the state and formation information, each firm has its own Normal return distribution, and returns are assumed independent across firms. At formation, the state is unknown, so each firm's forecast includes all three components.

Sharing a state allows the model to describe related movements in firms' returns. If several firms have lower expected returns in the same state, uncertainty about that state can link their returns. The key assumption is that firms face the same state realization, in addition to using the same state probabilities. The next subsection explains how this

assumption enters the training objective. Appendix B.6 derives the implied covariances and correlations.

Structured K3 lets macroeconomic conditions change state probabilities while firm information determines the return distribution within each state. FiLM Sigma and FiLM All allow a further possibility: the same firm characteristics may imply different risks under different macroeconomic conditions. They implement this idea by using macroeconomic information to adjust the learned firm features before predicting component parameters. Let $h_{i,t} = h(x_{i,t}, e_{i,t}) \in \mathbb{R}^8$ be the firm's eight learned features and $g_t = g(m_t) \in \mathbb{R}^8$ the eight learned macro features. The functions h and g are the firm and macro networks. FiLM stands for feature-wise linear modulation: each learned firm feature is multiplied by a macro-dependent factor and receives a macro-dependent additive adjustment. As Figure 2 shows, the macro features g_t produce two eight-dimensional vectors: a *scale change* $\Delta\gamma_t$ and a *shift* β_t:

$$\Delta\gamma_t = 0.1 \tanh\{a(g_t)\}, \qquad \beta_t = 0.1 \tanh\{b(g_t)\}. \tag{7}$$

The functions a and b are learned linear transformations with intercepts; tanh is applied element by element. The scale change $\Delta\gamma_t$ modifies the multiplier on each firm feature, while the shift β_t adds to that feature. Let u_k and s_k be the learned output functions for component k's mean and scale score. FiLM All uses the adjusted features to predict both component means and standard deviations:

$$\widetilde{h}_{i,t} = (1 + \Delta\gamma_t) \odot h_{i,t} + \beta_t, \tag{8}$$

$$\mu_{i,k,t} = u_k(\widetilde{h}_{i,t}), \tag{9}$$

$$\sigma_{i,k,t} = \text{softplus}\{s_k(\widetilde{h}_{i,t})\} + 0.005. \tag{10}$$

The operator $\odot$ denotes element-by-element multiplication. Softplus makes the standard deviation positive, and adding 0.005 sets its lower bound. Both $\Delta\gamma_t$ and β_t are shared by all firms in month t. The multiplicative adjustment depends on each firm's original features $h_{i,t}$; the additive shift is the same across firms.

FiLM Sigma uses the same feature adjustment and standard-deviation equation, but its mean function receives the original firm features: $\mu_{i,k,t} = u_k(h_{i,t})$. Thus macroeconomic conditions adjust the features used for component standard deviations, while the component means depend only on firm information. FiLM All allows the adjustment to affect both. All three specifications use a common state and macro-driven state probabilities. Even in FiLM Sigma, the overall predictive mean can change with macroeconomic conditions

through these probabilities, although the component means do not depend directly on macroeconomic inputs.

In both variants, bounding each entry of $\Delta\gamma_t$ and β_t between -0.1 and 0.1 limits the feature multiplier to between 0.9 and 1.1 and the absolute additive shift to less than 0.1.

Figure 2 shows how the three specifications use the same inputs. The macro features g_t determine the common probabilities in every model. In the FiLM models, they also produce the scale change and shift used to form adjusted firm features $\widetilde{h}_{i,t}$. The table inside the figure shows which features enter each output head. Structured K3 uses the original features $h_{i,t}$ for both means and scales. FiLM Sigma uses adjusted features only for scales, and FiLM All uses them for both.

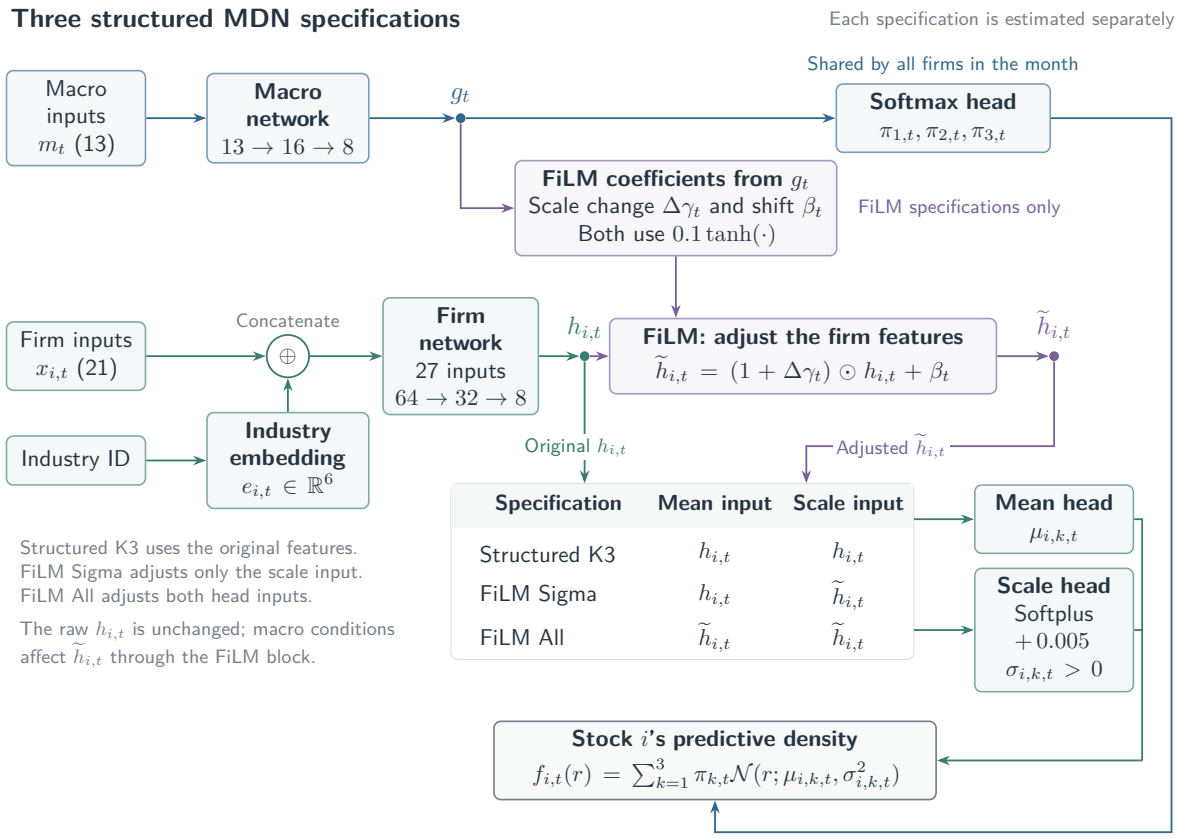


Figure 2: Three structured MDN specifications

Notes: Macro features g_t determine the common probabilities in every specification. In the FiLM specifications, they also generate the scale change $\Delta\gamma_t$ and shift β_t used to form $\widetilde{h}_{i,t}$ from the original firm features $h_{i,t}$. The input table specifies which features enter each head. The FiLM path is absent in Structured K3. All specifications retain three Normal components and are estimated separately. Feature multiplication is element by element; the same monthly adjustment vectors are applied to every firm.

2.2.3 Rolling Estimation and Training

Figure 3 shows how the estimation sample moves through time. Each window has 72 training months, followed by 12 validation months and 12 test months. In the first window, formation years 2000–2005 are used for training, 2006 for validation, and 2007 for testing. All three blocks then move forward by 12 months. The final window trains on 2017–2022, validates on 2023, and tests on 2024. This gives 18 test windows, indexed by $w = 0, \ldots, 17$. Dates refer to formation months: inputs in month t predict the return in month $t + 1$.

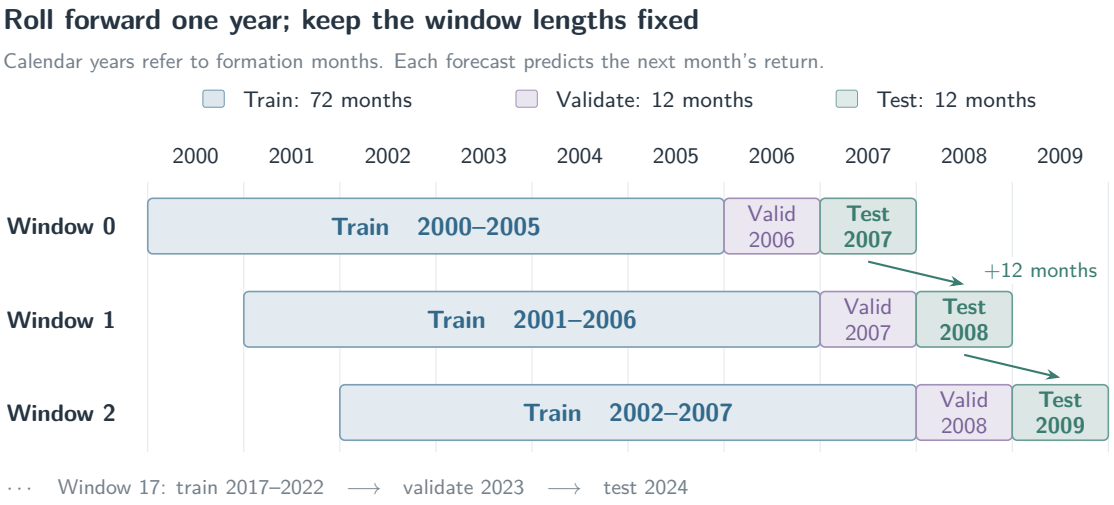


Figure 3: Annual rolling estimation schedule. Each window contains 72 training months, 12 validation months, and 12 test months. All three blocks advance by 12 months, giving 18 test windows indexed by $w = 0, \ldots, 17$ over formation years 2007–2024. The first three windows and the final window are shown. Dates refer to formation months: information in month t is used to forecast the return in month $t + 1$.

Figure 4 shows the procedure within each window: fit the model, select a checkpoint, assign component labels, and forecast returns. Every neural model starts from a new initialization. In the FiLM models, the coefficient layer is initialized at zero, giving $\Delta\gamma_t = \beta_t = 0$ and hence $\widetilde{h}_{i,t} = h_{i,t}$ at initialization. Early stopping selects the saved model with the lowest validation loss. The main training procedure does not refit the selected model on the combined training and validation data.

For the structured mixtures, the selected model is then evaluated on the 84 training and validation months to rank its component means. Let $\mathcal{A}_w$ be this set of months, with s indexing a formation month and N_s denoting its firm count. In Figure 4, $\bar{\mu}_{k,w}$ is component k's fitted mean averaged first across firms and then equally across $\mathcal{A}_w$. The index $A_w = \arg\min_k \bar{\mu}_{k,w}$ identifies the component labeled adverse; A_w is a component index, while $\mathcal{A}_w$ is a set of months. Section 4.2 defines the labels formally. This step

assigns labels without updating model parameters. Parameters and labels remain fixed during the next 12 test months, while forecasts change with monthly inputs. The model is fitted and the labels are assigned again in the next window.

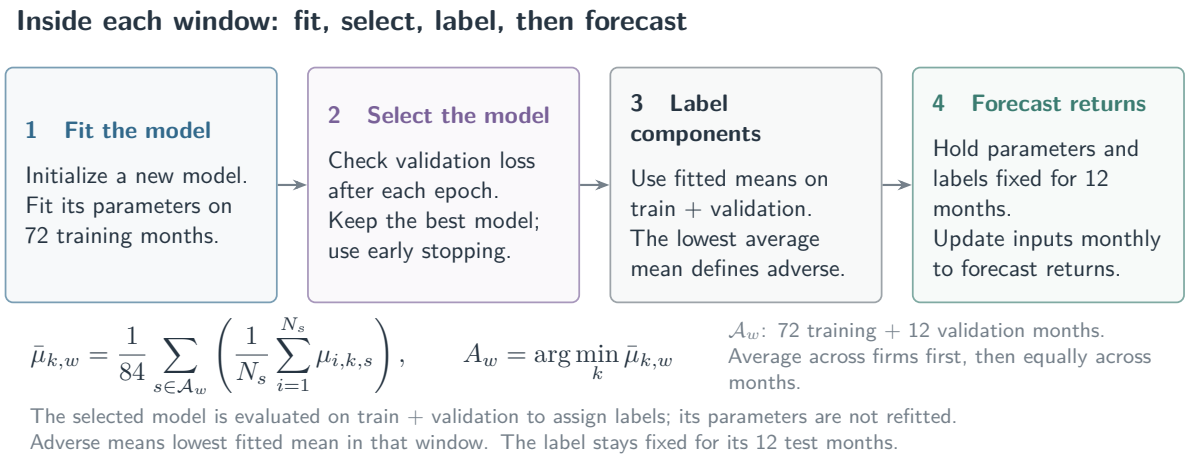


Figure 4: Estimation and forecasting within each window. A newly initialized model is fitted on the training sample, and validation loss selects the checkpoint. The selected model is evaluated on the 84 training and validation months to assign component labels, without refitting its parameters. Component means are averaged first across firms and then equally across months; the lowest-mean component is labeled adverse. Parameters and labels remain fixed during the next 12 test months, while forecasts vary with monthly inputs. The procedure is repeated in each estimation window.

For continuous firm predictor $j = 1, \ldots, 21$, let $x_{i,j,t}^{\text{imp}}$ be its cleaned and imputed value before scaling, as described in Section 2.1. Let $\overline{x}_j^{\text{tr},w}$ and $s_j^{\text{tr},w}$ be its pooled mean and sample standard deviation over the training observations in window w. The scaled predictor is

$$x_{i,j,t}^{\text{sc}} = \frac{x_{i,j,t}^{\text{imp}} - \overline{x}_j^{\text{tr},w}}{s_j^{\text{tr},w} + 10^{-8}}. \tag{11}$$

Collecting the 21 scaled values gives $x_{i,t} = (x_{i,1,t}^{\text{sc}}, \ldots, x_{i,21,t}^{\text{sc}})^{\mathsf{T}}$, the firm-input vector in equation (5).

The mean and sample standard deviation are calculated across all training stock–months in window w, rather than separately within each month. Validation and test observations use the same training-window quantities, without re-estimating them on later data. Macro variables are scaled using one observation per training month, and missing values use training medians. Return targets remain in their original units. These forecasting transformations are separate from the monthly transformations used in the later regressions.

The networks are trained together using next-month returns from complete monthly cross-sections. For each formation month t, let $\mathcal{U}_t$ be the set of firms in the estimation sample, and let $\boldsymbol{r} = (r_i)_{i \in \mathcal{U}_t}$ contain one possible return for each firm. The common-state model assigns this return vector the joint density

$$f_t(\boldsymbol{r}) = \sum_{k=1}^{3} \pi_{k,t} \prod_{i \in \mathcal{U}_t} \mathcal{N}(r_i; \mu_{i,k,t}, \sigma^2_{i,k,t}), \tag{12}$$

To evaluate the observed monthly return vector, I set $r_i = R_{i,t+1}$. Within each possible state, the model multiplies the firms' Normal densities, reflecting the assumption of conditional independence. It then averages these products using the state probabilities. The state is unobserved, so all three possibilities enter the likelihood. The network parameters that determine probabilities, means, and standard deviations are estimated jointly through the training objective below.

The structured training objective has two parts: a normalized joint negative log likelihood (NLL) and a marginal NLL. Figure 5 shows how both are calculated from the same model outputs. Write $f_{i,k,t}(r) = \mathcal{N}(r; \mu_{i,k,t}, \sigma^2_{i,k,t})$ for firm i's component-k density, before weighting by the common probability $\pi_{k,t}$. Let $\mathcal{T}_w$ be the set of 72 training months in window w, and let $N_t = |\mathcal{U}_t|$ be the number of firms in month t. The average number of firms per training month in window w is $\overline{N}_w = |\mathcal{T}_w|^{-1} \sum_{t \in \mathcal{T}_w} N_t$.

The left-hand path in the figure takes the negative log of the joint density in equation (12) and normalizes it by $\overline{N}_w$. The right-hand path evaluates each stock's marginal density separately and averages the individual negative log likelihoods across firms. The two monthly losses are

$$\ell^J_t = -\frac{1}{\overline{N}_w} \log \left[\sum_{k=1}^{3} \pi_{k,t} \prod_{i \in \mathcal{U}_t} f_{i,k,t}(R_{i,t+1}) \right], \tag{13}$$

$$\ell^M_t = -\frac{1}{N_t} \sum_{i \in \mathcal{U}_t} \log \left[\sum_{k=1}^{3} \pi_{k,t} f_{i,k,t}(R_{i,t+1}) \right]. \tag{14}$$

The superscripts J and M label the joint and marginal terms. "Marginal" means that each stock's distribution is evaluated separately, while the joint term evaluates the return vector under the common-state model. The joint NLL is divided by the average training-month firm count to put it on a comparable scale to the average individual-stock NLL. Both paths use every firm and all three states. The joint term combines all firms' returns within each state, then averages over the states using their common probabilities. The marginal path averages over states separately for each stock. Neither assigns a firm to a single

component.

I combine the two monthly losses and average equally across training months. The structured training objective is

$$\mathcal{L}_w^S = \frac{1}{|\mathcal{T}_w|} \sum_{t \in \mathcal{T}_w} \left(0.9\ell_t^J + 0.1\ell_t^M\right). \tag{15}$$

Here $\mathcal{L}_w^S$ is the training objective for window w, and the superscript S identifies the structured model. It places weight 0.9 on the normalized joint loss and 0.1 on the marginal loss, as shown in Figure 5. This combines evaluation of the common-state joint distribution with evaluation of individual return distributions. Validation uses the same combination and the training value of $\overline{N}_w$. Each training batch contains complete monthly cross-sections; no firms are subsampled. Appendix B gives the network and optimization settings.

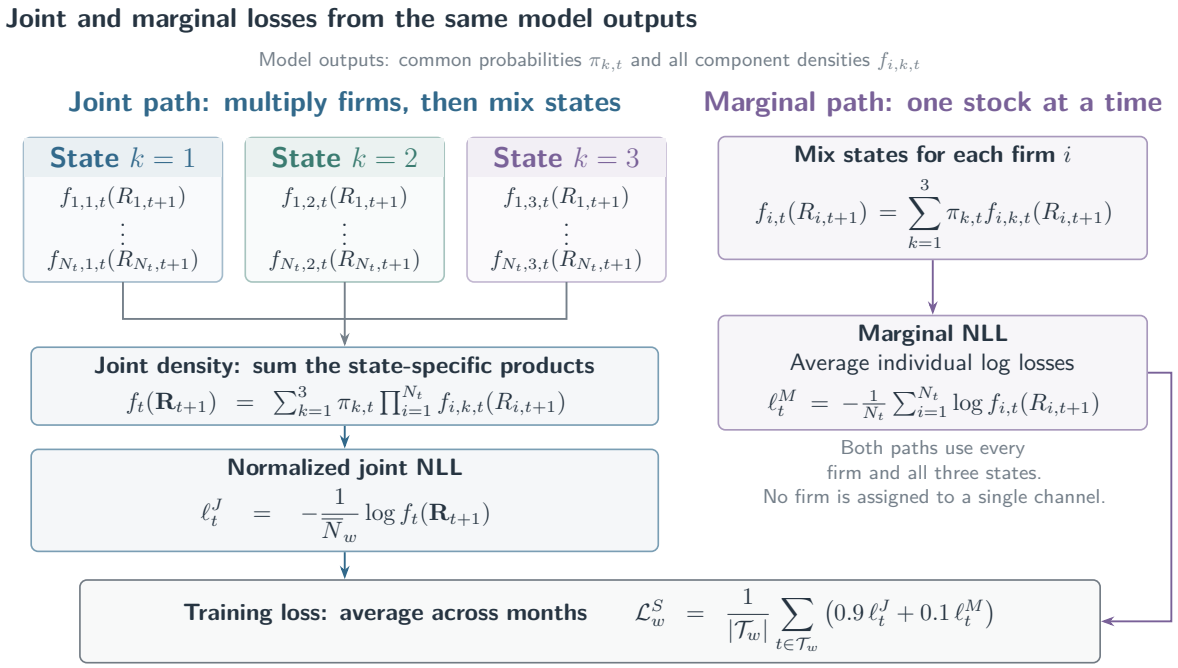


Figure 5: Common-state joint NLL and individual marginal NLL

Notes: The joint path multiplies all firms' component densities within each state and then mixes across states. The marginal path first mixes states for each firm and then averages individual negative log likelihoods. Here N_t is the monthly firm count and $\overline{N}_w$ is its training-window average, held fixed for validation. The structured objective combines the normalized joint and marginal terms with weights 0.9 and 0.1 and averages equally across months. Both paths retain every firm and all three states.

3 Model Comparison and Forecast Evaluation

I compare seven models: Rolling Normal, Linear Quantile, Fused K1, Fused K3, Structured K3, FiLM Sigma, and FiLM All. Rolling Normal uses each stock's past 84 calendar months of returns. Linear Quantile estimates return quantiles as linear functions of firm predictors, macroeconomic predictors, and industry information. Fused K1 uses a neural network but keeps a single Normal distribution. Fused K3 adds three mixture components without separating the roles of macro and firm inputs. The structured models assign distinct roles to the two input sets, and FiLM also lets macroeconomic conditions adjust learned firm features. Appendix B describes the benchmarks.

Table 3: Model designs and the role of each input set

Model	Probabilities	Component means	Component standard deviations
Rolling Normal	Single component	Own return history	Own return history
Linear Quantile	Not applicable	Not parameterized	Not parameterized
Fused K1	Single component	Fused macro and firm	Fused macro and firm
Fused K3	Fused macro and firm	Fused macro and firm	Fused macro and firm
Structured K3	Common macro probabilities	Firm features	Firm features
FiLM Sigma	Common macro probabilities	Firm features	Macro-adjusted firm features
FiLM All	Common macro probabilities	Macro-adjusted firm features	Macro-adjusted firm features

Notes: All neural density models use the same 21 firm variables and 13 macro variables, and each estimates its own six-dimensional industry embedding. Fused K1 has one Normal component; Fused K3 has three Normal components. The structured models use one common monthly state and the training loss in equation (15). Linear Quantile predicts grid quantiles directly rather than estimating component means or standard deviations.

3.1 Forecast Accuracy Metrics

I assess predictive accuracy by comparing the distributions forecast at formation with the returns that subsequently occur. Five loss functions evaluate two aspects of these forecasts. Negative log likelihood (NLL) and the continuous ranked probability score (CRPS) assess the full return distribution. The remaining metrics focus on the lower tail: the quantile score evaluates the VaR cutoff, lower-tail CRPS evaluates a range of lower-tail quantiles, and FZ0 evaluates VaR and ES jointly. For every metric, a lower average loss indicates greater predictive accuracy. Scores are compared across models within each metric.

For each stock i, I evaluate the forecast made at the end of month t using the stock's realized return in month $t+1$. Let $\ell_{i,t}$ denote the resulting loss under the metric being evaluated. I first average these losses across stocks within each month, then give each monthly average equal weight:

$$\frac{1}{T}\sum_{t}\left(\frac{1}{N_t}\sum_{i}\ell_{i,t}\right).$$

Here $T = 216$, and N_t is the number of stocks included in the calculation for month t. All models in a comparison are evaluated on the same stocks. This averaging gives each month equal importance even when the number of stocks changes. Statistical tests use the monthly loss differences between models, as described below.

3.1.1 Negative Log Likelihood

NLL measures how much density a model assigns to the return that actually occurs:

$$\mathrm{NLL}_{i,t} = -\log f_{i,t}(R_{i,t+1}). \tag{16}$$

A model receives a lower loss when it assigns a higher density at the realized return. Assigning very little density to an observed return produces a large penalty. Averaged over observations, NLL therefore evaluates the accuracy of the predictive density, including its ability to accommodate unusual returns. I report NLL for the six density models. Linear Quantile is excluded from this metric because its finite set of quantiles does not specify a unique density.

3.1.2 Continuous Ranked Probability Score

CRPS evaluates the cumulative probabilities across the full return distribution. At each possible return cutoff, it compares the forecast probability of falling below that cutoff with whether the realized return did so. It then integrates the squared differences across cutoffs. This complements NLL, which evaluates density at the realized return.

For a predictive CDF F and realized return y, CRPS is

$$\mathrm{CRPS}(F, y) = \int_{-\infty}^{\infty} [F(z) - \mathbf{1}\{y \le z\}]^2 \, dz \tag{17}$$

$$= 2\int_{0}^{1} \rho_u(y - Q(u)) \, du. \tag{18}$$

Here z is a return cutoff, $\mathbf{1}\{A\}$ equals one when condition A holds and zero otherwise,

and $Q = F^{-1}$ is the quantile function. In the second expression, u is a probability level and $\rho_u(v) = v[u - \mathbf{1}\{v < 0\}]$ is the pinball loss. This equivalent expression shows that CRPS also averages quantile losses over the full probability range. A lower average CRPS indicates a more accurate full-distribution forecast (Gneiting and Raftery, 2007).

For the seven-model comparison, I calculate CRPS and lower-tail CRPS on the same grid of 27 probability levels using a common numerical integration rule. This allows Linear Quantile to be evaluated alongside the density models. Appendix B.4 gives the calculation details.

3.1.3 Quantile Score

The quantile score evaluates the forecast cutoff at one probability level. At $\alpha = 0.05$, for example, it assesses the predicted fifth percentile of next month's return. Using the pinball loss defined above, the score is

$$\mathrm{QS}_{i,t,\alpha} = 2\rho_\alpha(R_{i,t+1} - q_{i,t,\alpha})\,. \tag{19}$$

The loss penalizes errors asymmetrically: at a low probability level, a return below the forecast cutoff receives a larger penalty than an equally distant return above it. The correct conditional quantile minimizes the expected quantile loss (Koenker and Bassett, 1978; Gneiting, 2011). Lower average QS therefore indicates a more accurate VaR forecast. I report it at both the 5% and 10% levels.

3.1.4 Lower-Tail CRPS

Whereas QS evaluates one cutoff, lower-tail CRPS evaluates all quantiles from zero to the selected tail probability α:

$$\mathrm{TailCRPS}_{i,t}^{0:\alpha} = \frac{2}{\alpha}\int_0^\alpha \rho_u(R_{i,t+1} - Q_{i,t}(u))\; du. \tag{20}$$

The superscript $0:\alpha$ identifies this probability range. Dividing by α expresses the score as an average over the range. At the 5% level, the score evaluates the most extreme lower-tail quantiles; at 10%, it covers a wider part of the lower tail. Lower values indicate more accurate quantile forecasts across that range (Gneiting and Ranjan, 2011). Reporting both QS and lower-tail CRPS distinguishes accuracy at the VaR cutoff from accuracy throughout the tail below it.

3.1.5 Joint VaR–ES Score

The final metric evaluates the VaR cutoff and the average return below it together. For one stock and formation month, write $y = R_{i,t+1}$, $q = q_{i,t,\alpha}$, and $e = \mathrm{ES}_{i,t,\alpha}$. I use the FZ0 loss specified by Patton et al. (2019), within the class of joint VaR–ES scores characterized by Fissler and Ziegel (2016):

$$L_{FZ0}(y, q, e; \alpha) = -\frac{\mathbf{1}\{y \le q\}(q - y)}{\alpha e} + \frac{q}{e} + \log(-e) - 1. \tag{21}$$

A lower average FZ0 indicates more accurate joint VaR–ES forecasts. It complements the quantile scores by also evaluating the ES forecast; it does not measure ES accuracy in isolation. I report FZ0 at both the 5% and 10% levels.

ES retains its signed return units. The logarithm requires $e < 0$. For the reported comparisons, I also require $q < 0$ and $e \le q$. At each tail level, I retain only observations for which all seven models satisfy these conditions, so every model is evaluated on the same stocks. Appendix B.4 reports how many observations are retained.

3.1.6 Out-of-Sample Forecast Performance

Table 4: Out-of-sample distribution and tail forecast losses

Model	NLL	CRPS	Tail 5%	Tail 10%	QS 5%	QS 10%	FZ0 5%	FZ0 10%
Rolling Normal	61486.5495$^{(6)}$	0.0735$^{(4)}$	0.0179$^{(6)}$	0.0286$^{(5)}$	0.0301$^{(5)}$	0.0478$^{(4)}$	-1.2190$^{(4)}$	-1.4818$^{(4)}$
Linear Quantile	—	0.0863$^{(7)}$	0.0211$^{(7)}$	0.0361$^{(7)}$	0.0375$^{(7)}$	0.0623$^{(7)}$	-1.1736$^{(7)}$	-1.3987$^{(6)}$
Fused K1	-0.5029$^{(5)}$	0.0780$^{(5)}$	0.0170$^{(4)}$	0.0284$^{(4)}$	0.0297$^{(4)}$	0.0490$^{(5)}$	-1.2173$^{(5)}$	-1.4077$^{(5)}$
Fused K3	-0.5051$^{(4)}$	0.0813$^{(6)}$	0.0176$^{(5)}$	0.0293$^{(6)}$	0.0308$^{(6)}$	0.0503$^{(6)}$	-1.1774$^{(6)}$	-1.3606$^{(7)}$
Structured K3	-0.6406$^{(3)}$	0.0722$^{(3)}$	0.0159$^{(2)}$	0.0265$^{(2)}$	0.0278$^{(2)}$	0.0457$^{(3)}$	-1.3648$^{(2)}$	-1.5633$^{(2)}$
FiLM Sigma	-0.6608$^{(1)}$	0.0712$^{(1)}$	0.0158$^{(1)}$	0.0263$^{(1)}$	0.0277$^{(1)}$	0.0453$^{(1)}$	-1.3701$^{(1)}$	-1.5653$^{(1)}$
FiLM All	-0.6475$^{(2)}$	0.0717$^{(2)}$	0.0159$^{(3)}$	0.0265$^{(3)}$	0.0279$^{(3)}$	0.0456$^{(2)}$	-1.3516$^{(3)}$	-1.5489$^{(3)}$

Notes: Formation dates run from January 2007 through December 2024: 216 months and 480,446 common stock–months. Lower scores mean better forecasts. CRPS and tail CRPS use the same quantile grid for all models. FZ0 uses observations where all models have valid negative-ES forecasts. Linear Quantile has no exact density, so NLL is not reported. Rolling Normal estimates volatility with as few as two observed monthly returns. Nearly identical returns can produce a very small variance, so a later return away from the estimated mean receives extremely low density and generates a large NLL penalty. Sections 3.1.1–3.1.5 define the scores. Superscripts in parentheses report ranks within each score column (1 is best), computed from unrounded losses among models with that score available. Exact ties share a rank; rounded values can be identical despite different ranks.

Table 4 shows that Structured K3, FiLM Sigma, and FiLM All occupy the top three positions in every score column. Their advantage spans the full return distribution and the lower tail, with each structured specification outperforming the historical Normal, linear

quantile, and fused neural benchmarks wherever the score is available.

For the full distribution, the structured models' NLL values range from −0.6608 to −0.6406, compared with −0.5029 for Fused K1 and −0.5051 for Fused K3. Their CRPS values are 0.07123–0.07222, compared with 0.07803 and 0.08132. These represent CRPS reductions of approximately 7.4–8.7% relative to Fused K1 and 11.2–12.4% relative to Fused K3. The improvement is therefore visible both in the density assigned to realized returns and in cumulative probabilities across the distribution.

The tail forecasts improve as well. All three structured models have lower QS and tail CRPS at both probability levels, indicating more accurate VaR cutoffs and lower-tail quantiles. Their joint VaR–ES forecasts also have lower FZ0 losses. At 5%, FZ0 ranges from −1.3701 to −1.3516, compared with −1.2173 for Fused K1 and −1.1774 for Fused K3. At 10%, the corresponding range is −1.5653 to −1.5489, compared with −1.4077 and −1.3606. Thus the gains extend to the tail measures used in the subsequent economic analysis.

To assess statistical significance, I compare each structured model with both fused neural benchmarks. These benchmarks use the same firm, macroeconomic, and industry information, making them the main comparison for evaluating the structured forecasting procedures. Let a denote a structured model and b a benchmark. Their monthly average loss difference on the same stocks is

$$d_t^{a,b} = \frac{1}{N_t} \sum_i \left[\ell_{i,t}^a - \ell_{i,t}^b \right]. \tag{22}$$

A negative average difference favors the structured model. I test the null hypothesis of equal expected loss, $\mathbb{E}[d_t^{a,b}] = 0$, using the monthly differences (Diebold and Mariano, 1995). I use Newey–West standard errors with 12 calendar-month lags (NW12) (Newey and West, 1987). The reported p-values are two-sided.

Table 5 confirms that the gains are statistically significant across all eight score columns. Panel A reports NLL and CRPS, together with FZ0 at the 5% and 10% levels; Panel B reports QS and tail CRPS at both levels. Across three structured models, two fused benchmarks, and eight scores, all 48 loss differences are negative and significant at 5%. This evidence covers density accuracy, the full CDF, the VaR cutoffs, the range of lower-tail quantiles, and joint VaR–ES forecasts. Figure 6 illustrates the CRPS and FZ0 differences relative to Fused K3.

Table 5: Paired tests for all eight forecast scores

Panel A. Full-distribution and joint VaR–ES scores									
Candidate	Benchmark	Δ NLL	p	Δ CRPS	p	Δ FZ0 5%	p	Δ FZ0 10%	p
Structured K3	Fused K1	-0.1377	< 0.001	-0.0058	0.005	-0.1474	< 0.001	-0.1556	< 0.001
Structured K3	Fused K3	-0.1355	0.006	-0.0091	0.004	-0.1873	< 0.001	-0.2027	< 0.001
FiLM Sigma	Fused K1	-0.1579	< 0.001	-0.0068	0.001	-0.1528	< 0.001	-0.1577	< 0.001
FiLM Sigma	Fused K3	-0.1557	0.004	-0.0101	0.003	-0.1927	< 0.001	-0.2047	< 0.001
FiLM All	Fused K1	-0.1446	< 0.001	-0.0064	0.002	-0.1343	< 0.001	-0.1412	< 0.001
FiLM All	Fused K3	-0.1423	0.008	-0.0097	0.004	-0.1742	< 0.001	-0.1883	< 0.001

Panel B. Quantile and lower-tail distribution scores									
Candidate	Benchmark	Δ QS 5%	p	Δ QS 10%	p	Δ Tail 5%	p	Δ Tail 10%	p
Structured K3	Fused K1	-0.0019	0.003	-0.0033	0.004	-0.0011	0.003	-0.0019	0.003
Structured K3	Fused K3	-0.0030	< 0.001	-0.0046	0.001	-0.0017	< 0.001	-0.0028	< 0.001
FiLM Sigma	Fused K1	-0.0021	0.001	-0.0037	0.002	-0.0012	0.001	-0.0020	0.001
FiLM Sigma	Fused K3	-0.0031	< 0.001	-0.0050	0.002	-0.0018	< 0.001	-0.0029	< 0.001
FiLM All	Fused K1	-0.0019	0.009	-0.0034	0.005	-0.0010	0.019	-0.0018	0.007
FiLM All	Fused K3	-0.0029	0.002	-0.0047	0.003	-0.0016	0.002	-0.0027	0.002

Notes: Each entry Δ is the structured model's average loss minus the fused benchmark's average loss. Negative differences favor the structured model. Panel A reports NLL and CRPS, together with FZ0 at 5% and 10%; Panel B reports QS and tail CRPS at both levels. All three structured models are compared with both fused benchmarks on every score. Losses are averaged across stocks within each month, then equally across the 216 formation months. Each FZ0 comparison uses the common valid sample at its tail level. Two-sided p-values use Newey–West standard errors with 12 calendar-month lags (NW12).

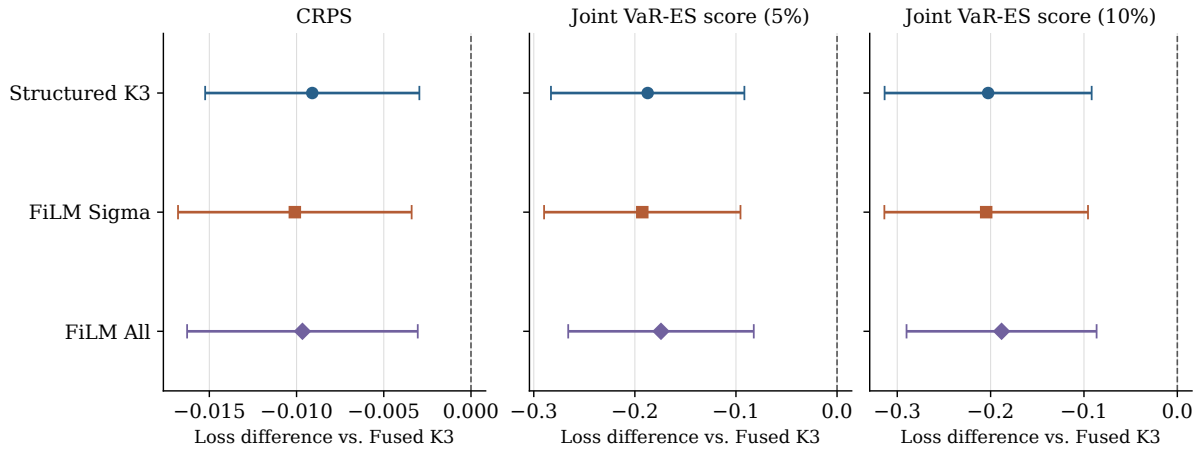


Figure 6: Predictive-accuracy gains relative to Fused K3

Notes: Points show each model's average loss minus Fused K3 loss over 216 formation months. Lines show pointwise 95% confidence intervals based on NW12 standard errors. Negative values favor the structured model. The two FZ0 panels use their respective common valid samples at the 5% and 10% tail levels.

The structured MDN framework thus improves out-of-sample forecasts of both the full return distribution and downside risk relative to the fused neural benchmarks. The advantage is shared by all three structured specifications: it holds both when macroeconomic information determines only the common state probabilities and when it also adjusts learned firm features. The evidence therefore supports a framework that separates the common market environment from firms' return distributions within that environment and estimates them jointly.

3.2 Probability Integral Transform

The forecast scores measure how accurately the models predict realized returns. I next examine whether their predicted probabilities agree with the frequencies of realized outcomes. For a continuous forecast, the probability integral transform (PIT) is

$$U_{i,t} = F_{i,t}(R_{i,t+1}).$$

It records the position of the realized return within the predicted distribution. If the conditional distribution is correct, PIT values are uniformly distributed between zero and one (Diebold et al., 1998; Gneiting et al., 2007). Dividing this range into ten equal bins therefore gives a reference frequency of 10% in each bin.

I calculate bin frequencies within each month and then average equally across the 216 months. Table 6 also reports monthly PIT means, variances, and Kolmogorov–Smirnov (KS) distances averaged across months, together with the first-order autocorrelation of the monthly PIT mean, ACF(1). A smaller KS distance indicates a distribution closer to uniformity. Density models use their exact CDFs; Linear Quantile uses interpolation between its sorted quantiles. Common market shocks can shift many stocks' PIT values in the same direction. Even a correctly specified model therefore need not produce a uniform cross-sectional PIT distribution in every month. I assess calibration over the full evaluation period and use the monthly summaries to compare models.

Figure 7 shows that the structured models produce broadly even PIT distributions, with particularly flat profiles for the two FiLM variants. Bin frequencies range from 8.6% to 11.4% for FiLM Sigma and from 8.7% to 11.2% for FiLM All, close to the 10% reference. Structured K3 also remains comparatively balanced, with frequencies between 8.2% and 12.3%. The fused models show much larger concentrations in the highest PIT bin: 15.0% for Fused K1 and 18.8% for Fused K3. The structured models substantially reduce this concentration and distribute realized outcomes more evenly across predicted probabilities.

The monthly diagnostics support this visual comparison. The structured models'

Table 6: PIT diagnostics and VaR coverage at both tail levels

Model	PIT mean	Variance	KS	ACF(1)	Cov. 5%	p	Cov. 10%	p
Rolling Normal	0.4827	0.0597	0.1985	-0.0511	5.393	0.640	9.409	0.591
Linear Quantile	0.5253	0.0660	0.2481	0.0458	6.353	0.294	11.508	0.407
Fused K1	0.5461	0.0585	0.2467	0.2622	5.032	0.969	8.737	0.299
Fused K3	0.5490	0.0662	0.2572	0.2558	6.679	0.109	11.524	0.332
Structured K3	0.5259	0.0621	0.2074	-0.0049	4.847	0.806	8.764	0.175
FiLM Sigma	0.5091	0.0605	0.2023	-0.0544	4.748	0.650	8.789	0.137
FiLM All	0.5091	0.0610	0.2043	-0.0272	4.905	0.869	9.035	0.258

Notes: A uniform PIT has mean 0.5 and variance 1/12. Variance reports the average within-month cross-sectional variance, which need not equal 1/12 even under correct marginal forecasts when stocks share a common state. KS is the mean monthly cross-sectional distance from uniformity; ACF(1) uses monthly PIT means. Coverage entries are equal-month average exceedance percentages, tested against their respective 5% or 10% targets with NW12 standard errors. These are unconditional mean-coverage tests.

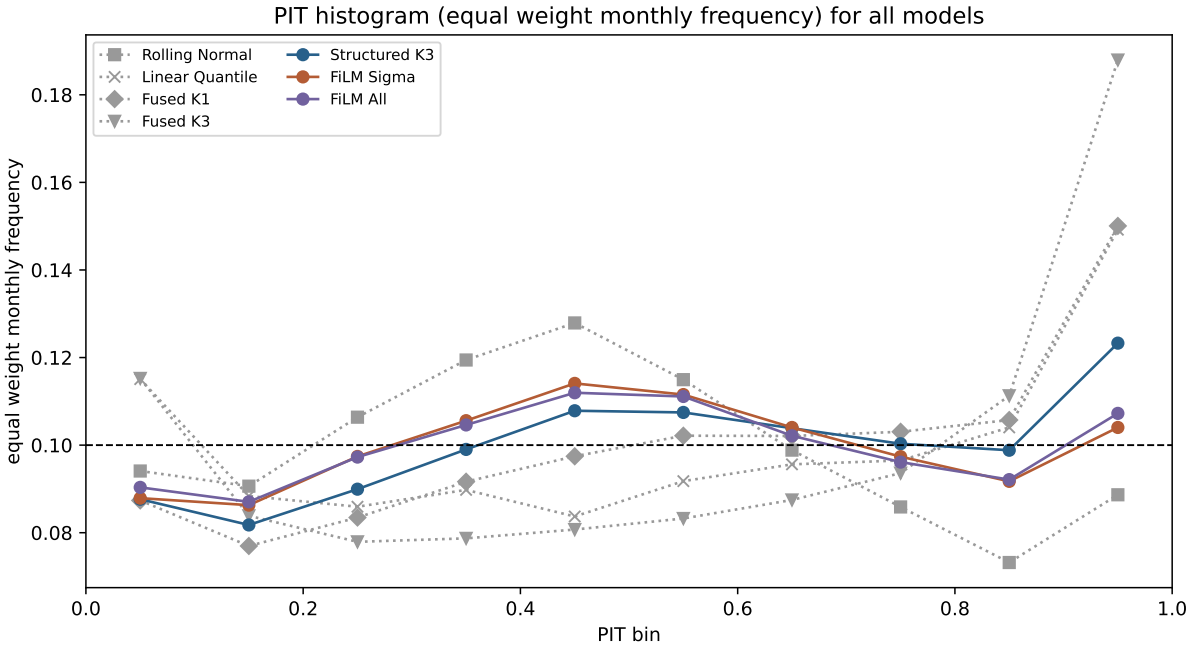


Figure 7: PIT distributions across models

Notes: Frequencies in ten equal-width PIT bins are calculated within each formation month and then averaged equally over the 216 months. The dashed horizontal line marks the uniform reference of 10% per bin. Solid colored curves show the three structured models; dotted gray curves show the benchmarks. Linear Quantile uses interpolation between its sorted quantiles.

average PIT values, 0.5091–0.5259, are closer to 0.5 than the fused values of 0.5461 and 0.5490. Their average KS distances are also lower, at 0.2023–0.2074 versus 0.2467 and 0.2572. Paired comparisons of monthly KS distances show that each structured model improves on both fused benchmarks, with $p < 0.001$ in all six comparisons. Rolling Normal has a slightly smaller average KS distance of 0.1985, but its differences from the structured models are not statistically significant: the three two-sided p-values range from 0.163 to 0.370. Appendix B.8 reports these tests. Thus, the structured models have smaller average monthly PIT KS distances than the fused neural benchmarks.

The structured models also show little persistence in their monthly average PIT deviations. ACF(1) ranges from -0.054 to -0.005, compared with approximately 0.26 for both fused models. Their monthly average PIT values therefore exhibit less first-order serial correlation. This complements the balanced histograms and lower KS distances.

I assess lower-tail calibration through unconditional VaR coverage: the average fraction of returns below the forecast cutoff should equal its nominal probability (Christoffersen, 1998). I test this condition using NW12 standard errors. The structured models' exceedance rates are 4.748%–4.905% at the 5% level and 8.764%–9.035% at the 10% level. NW12 tests do not reject the null of correct average coverage for any of the three models at either level; all six p-values exceed 0.13. These results are consistent with the stated tail probabilities in average coverage. They evaluate this particular calibration condition, separately from the descriptive PIT profiles and the comparisons of KS distances.

The structured models combine more accurate density and tail forecasts with balanced PIT profiles and lower average monthly KS distances than the fused benchmarks.

3.3 Overlap in ES-Based Stock Selection

The ES forecasts determine which stocks are classified as having the greatest or smallest predicted tail losses. I compare stock selections across all seven models to assess whether the three structured models identify similar stocks and how their selections differ from those of the benchmarks.

Each month, I sort stocks by each model's predicted ES and form two groups: the 10% with the lowest ES and the 10% with the highest ES. For each group, I measure the overlap in selected stocks between every pair of the seven models.

For models a and b, let $\mathcal{D}^a_{g,t}$ and $\mathcal{D}^b_{g,t}$ be their selected stock sets for extreme group g in month t. I measure their overlap by

$$O^{a,b}_{g,t} = \frac{2|\mathcal{D}^a_{g,t} \cap \mathcal{D}^b_{g,t}|}{|\mathcal{D}^a_{g,t}| + |\mathcal{D}^b_{g,t}|}. \tag{23}$$

Here g denotes the bottom or top 10% group, and $|\mathcal{D}^a_{g,t} \cap \mathcal{D}^b_{g,t}|$ counts the stocks selected by both models. The overlap measure divides this count by the average number of stocks selected by the two models. For example, if each model selects 100 stocks and 80 are shared, the overlap is 80%. It is 100% when the selections are identical and zero when no stocks are shared. I calculate overlap separately using 5% ES and 10% ES and report the average over 216 months. Two models selecting stocks independently at random would have an expected overlap of approximately 10%.

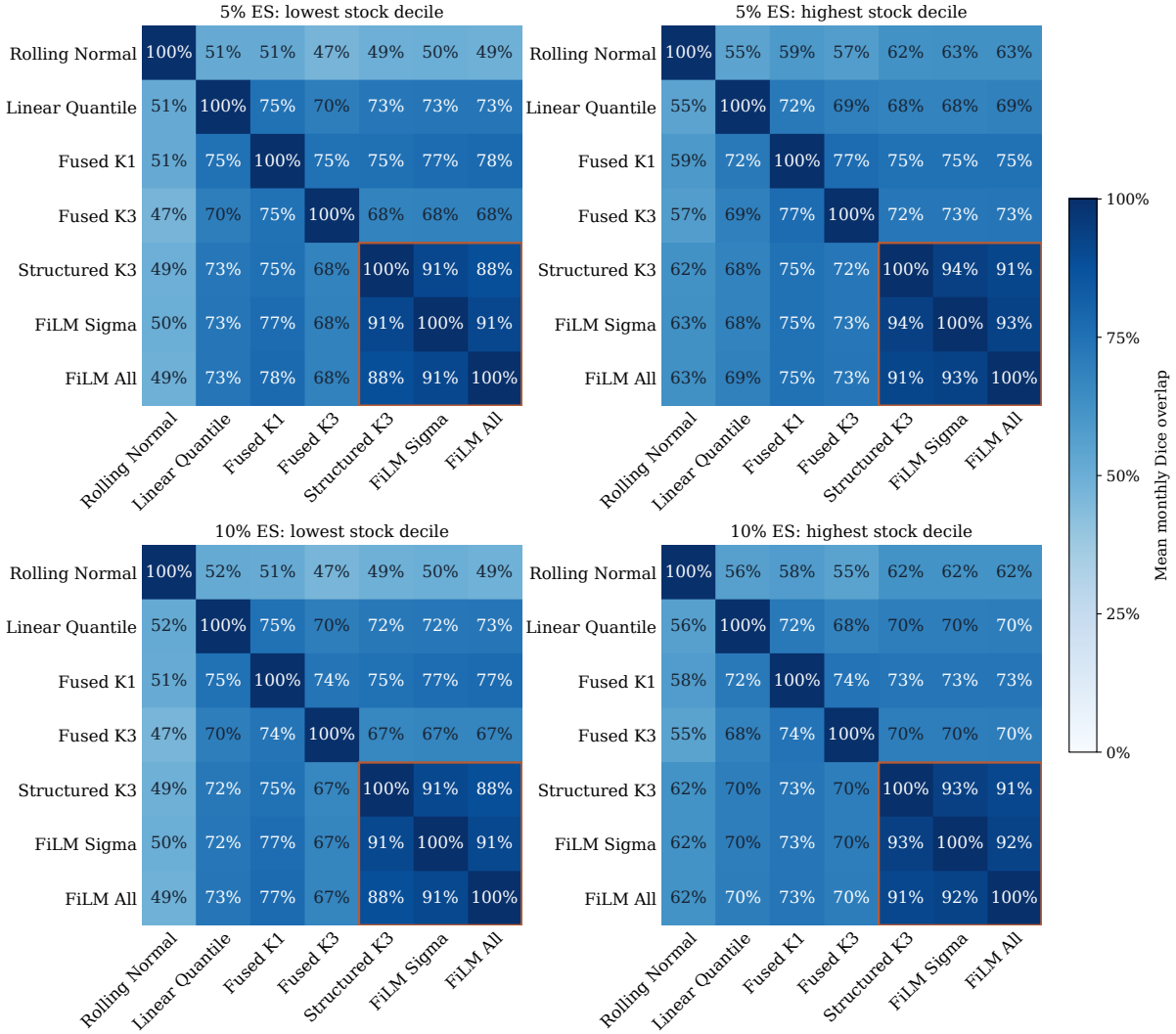


Figure 8: Overlap in extreme-ES stock selections

Notes: The matrices report average monthly overlap for every pair of the seven models, separately for the bottom and top 10% of stocks ranked by predicted ES. The top row uses 5% ES and the bottom row uses 10% ES. All models use the same seven-model forecast universe of 480,446 stock–months over 216 months. This sample precedes the additional portfolio eligibility restrictions. The outlined block contains the three structured specifications. The diagonal is 100% by construction. Overlap describes selected stock membership, not equality of predicted ES values.

Figure 8 shows that the three structured models agree much more closely with one another than any other pair of models in this seven-model comparison. At 5% ES, their lowest-decile overlaps range from 88.2% to 90.7%; at 10% ES, they range from 88.0% to 90.6%. No other model pair exceeds 77.6% and 77.2%, respectively. The same separation appears in the highest stock decile: within-family overlaps are 91.2%–93.7% at 5% ES and 90.8%–93.4% at 10% ES, while the largest overlaps among all remaining model pairs are 76.8% and 74.1%. Across both groups and both ES levels, overlap between any two structured models is at least 88%, whereas overlap for every other pair of models is below 78%.

The three structured specifications select largely the same stocks at both extremes of predicted tail risk, despite their different mean and scale parameterizations. Switching to another model changes stock selection much more than switching between structured variants. This evidence supports treating the three structured models as related implementations of the same design.

3.4 Mixture versus Normal Forecasts

The structured models use mixtures to allow asymmetry and differences in tail shape that a single Normal distribution cannot represent, features that can matter for asset pricing (Harvey and Siddique, 2000; Dittmar, 2002). I test whether this flexibility improves forecast accuracy when predicted mean returns and volatility are held fixed. For each forecast from Structured K3, FiLM Sigma, and FiLM All, I construct a Normal distribution with exactly the same mean and variance:

$$\widetilde{F}^{N}_{i,t}(r) = \Phi\left(\frac{r - \widehat{m}_{i,t}}{\widehat{v}_{i,t}}\right), \tag{24}$$

Here $\widehat{m}_{i,t}$ and $\widehat{v}_{i,t}$ are the mixture forecast's mean and standard deviation, calculated using equations (2) and (3). No additional model is trained. Since both forecasts have the same mean and variance, any difference in accuracy comes from the probabilities they assign to returns. Fused K1 predicts different means and volatilities, so its comparison with the structured models cannot show whether their better forecasts come from using a mixture.

I compare the forecasts using NLL, CRPS, and the joint VaR–ES score (FZ0) at the 5% and 10% levels. All comparisons use the same 480,446 stock–months over the full 216-month evaluation period. CRPS is calculated analytically for both distributions. I report each mixture's average loss minus that of its Normal counterpart, so negative differences favor the mixture. Statistical tests use monthly loss differences and Newey–West

standard errors with 12 calendar-month lags.

Table 7: Mixture and Normal forecasts with the same mean and variance

Model	Score	Difference	p
Structured K3	NLL	-0.012954	< 0.001
Structured K3	CRPS	0.000002	0.915
Structured K3	FZ0 5%	-0.000460	0.756
Structured K3	FZ0 10%	-0.000127	0.863
FiLM Sigma	NLL	-0.011006	< 0.001
FiLM Sigma	CRPS	-0.000020	0.378
FiLM Sigma	FZ0 5%	-0.000769	0.545
FiLM Sigma	FZ0 10%	-0.000467	0.512
FiLM All	NLL	-0.011405	0.007
FiLM All	CRPS	-0.000001	0.975
FiLM All	FZ0 5%	-0.002680	0.276
FiLM All	FZ0 10%	-0.001445	0.311

Notes: Each mixture is compared with a Normal distribution that has the same mean and variance. Differences are mixture loss minus Normal loss; negative values favor the mixture. All scores use the same 480,446 stock–months over the full 216-month evaluation period. CRPS is calculated analytically. Scores are averaged across stocks within each month and then across months. Two-sided p-values use Newey–West standard errors with 12 calendar-month lags.

Table 7 shows that all three mixtures have significantly lower NLL than their Normal counterparts. The differences are -0.01295 for Structured K3, -0.01101 for FiLM Sigma, and -0.01141 for FiLM All, with all three significant at 1%. Because each pair has identical means and variances, these gains show that using a mixture improves density forecasts beyond predicting mean returns and volatility. CRPS and FZ0 differences are not significant at 5%.

The Normal forecasts here also depend on the structured mixture architecture: their means and variances are calculated from its fitted components. The insignificant CRPS and FZ0 differences therefore concern the additional benefit of retaining the full mixture after using the architecture's predictions of mean returns and volatility. They do not overturn the structured models' gains over the fused benchmarks. The lower NLL shows that retaining the full mixture provides a further improvement in density forecasts.

3.5 Training Diagnostics

I examine the training records to assess whether the structured models continue to improve on validation data and whether their performance changes smoothly during training. I compare the five neural models using validation marginal NLL, which measures the

accuracy of individual-stock density forecasts and is recorded for all five. Lower values indicate better predictions on the held-out validation months. Appendix E.1 reports the training curves and calculation details.

The structured models benefit from continued training. In the main run, by epoch 20 their validation NLL is about 0.19 lower than after epoch 1, measured by the median change across the 18 rolling windows. Over the same interval, validation NLL rises by about 0.08–0.09 for the two fused benchmarks. Consistent with this pattern, Fused K1 and Fused K3 retain the model from epoch 1 in 10 and 11 of the 18 windows, respectively. None of the three structured models selects epoch 1 in the main run. Their median selected epochs range from 21.5 to 30, compared with epoch 1 for both fused models. These selections follow each model's own validation objective.

The structured models also show smaller changes between successive training epochs. I summarize these changes using the median absolute change in validation NLL between consecutive epochs over epochs 6–20, then take the median across windows. In the main run, this measure ranges from 0.0059 to 0.0084 for the structured models, compared with 0.0223–0.0236 for the fused benchmarks. The structured models therefore show smaller changes between epochs alongside continued improvement in validation accuracy.

These results complement the forecast comparisons. The structured design estimates common monthly state probabilities and firm-specific return distributions within each state, and its joint loss evaluates firms together within each month. In the training records, this procedure produces continued validation improvement and smaller changes between epochs. Together with the lower out-of-sample forecast losses, the evidence supports the structured models as effective forecasting procedures across rolling windows.

3.6 Main Forecast Results

All three structured models improve on both fused benchmarks across the eight forecast scores, with all 48 paired differences significant at 5%. The gains cover density forecasts, the full return distribution, and lower-tail risk. Their PIT profiles are more even than those of the fused benchmarks, and tests do not reject correct average VaR coverage at either tail probability level.

The structured variants select largely the same stocks at the extremes of predicted ES. Their training records also show continued validation improvement and smaller changes between epochs. Finally, the mixture–Normal comparison shows that retaining the mixture improves density forecasts even when the predicted mean and volatility are held fixed. These results support using the structured forecasts to examine stock downside risk.

4 Tail Risk and Stock Losses in Bad Markets

The structured forecasts provide economic information about both the size and the composition of a stock's tail risk. Total ES ranks stocks by the severity of their predicted losses. The component contributions show how much of that ES comes from each common state. This distinction matters when stocks have similar total ES but different exposure to the adverse component.

I first evaluate whether total-ES rankings distinguish stock performance in bad market months. I then test whether adverse contribution identifies additional crash vulnerability after controlling for total ES, historical risk exposures, and the model's predicted mean and volatility. Finally, stock groups formed from contribution residuals show the size of this difference in realized crash rates. The central finding is that tail composition adds economically meaningful information beyond the overall level of predicted tail risk.

All forecasts, stock rankings, and portfolio weights are determined before the next month's returns are observed. After observing next month's realized market excess return, I use the bad-market indicator to classify the month and evaluate the stocks selected beforehand.

In the main analysis, I define bad market months as the 20% of evaluation months with the lowest realized market excess returns. All remaining months are called other months. Let $M^e_{t+1} = M_{t+1} - R_{f,t+1}$, where M_{t+1} is the market return and $R_{f,t+1}$ is the risk-free return. The bad-market indicator B_{t+1} equals one in bad market months and zero in other months:

$$B_{t+1} = \mathbf{1}\{M^e_{t+1} \leq c_{.20}\}, \qquad c_{.20} = -0.0255. \tag{25}$$

Here $c_{.20}$ is the 20th percentile of realized market excess returns over the evaluation period. The cutoff identifies 44 bad market months, with an average market excess return of -6.11%, and 172 other months. Other months can still have negative market excess returns above this cutoff. This classification is separate from the model's fitted states and does not enter portfolio formation.

4.1 Total ES and Downside Performance

I test whether ES forecasts made at the end of month t identify stocks that perform worse if month $t+1$ is subsequently classified as a bad market month. Stocks are grouped using information available at t, and their performance is measured using realized returns in $t+1$. The test evaluates the stock ranking, without assuming that next month's market

conditions are known at formation.

4.1.1 Portfolio Formation and Evaluation

At the end of each month t, I sort eligible stocks separately on their 5% and 10% ES forecasts and form ten groups. Decile 1 contains the stocks with the lowest, most negative ES and greatest predicted tail losses; decile 10 contains those with the highest ES and smallest predicted tail losses. I hold these groups over month $t+1$. Using their realized returns, I calculate the return spread from buying decile 10 and selling decile 1:

$$H_{t+1}^{ES} = R_{10,t+1}^{ES} - R_{1,t+1}^{ES}. \tag{26}$$

Here $R_{d,t+1}^{ES}$ is the realized return in month $t+1$ on the stocks assigned to decile d at the end of t. I omit the ES probability level from the return notation.

I calculate both equal-weighted (EW) returns and value-weighted (VW) returns, with value weights based on formation-month market capitalization. Both sorts use the same 480,429 eligible stock–months. Stock groups and weights are fixed at formation, and each portfolio is held for one month. Returns are reported as monthly percentages before trading and stock-borrowing costs. Probability differences are reported in percentage points (pp).

After month $t+1$ ends, its realized market excess return determines whether it is a bad market month under equation (25). I then average the realized portfolio spreads over all months and separately over the 44 bad months and 172 other months. This classification is used only to evaluate portfolios already formed at t. Tests use Newey–West standard errors with 12 calendar-month lags (NW12), retaining the actual calendar spacing between bad months.

4.1.2 Returns in Bad and Other Months

Table 8 shows that stocks with less severe predicted tail losses perform substantially better in bad market months. At 5% ES, the equal-weighted long–short spreads earn 6.18%, 6.08%, and 5.96% per bad month for Structured K3, FiLM Sigma, and FiLM All. At 10% ES, the corresponding returns are 6.23%, 6.10%, and 5.91%. The value-weighted spreads are also large, ranging from 5.11% to 6.11% across the six model–ES combinations. All 12 bad-month spreads are positive with $p < 0.001$.

In the other 172 months, I find no significant difference in average returns between the highest- and lowest-ES groups. The high-minus-low spreads are negative: -0.48% to -0.41% with equal weights and -0.89% to -0.75% with value weights. These estimates

Table 8: ES portfolio returns across realized market conditions

Model	Weight	All months	p	Bad months	p	Other months	p
Panel A: 5% ES							
Structured K3	EW	0.883	0.113	6.184	< 0.001	-0.474	0.490
Structured K3	VW	0.535	0.344	5.887	< 0.001	-0.834	0.242
FiLM Sigma	EW	0.914	0.104	6.080	< 0.001	-0.408	0.550
FiLM Sigma	VW	0.429	0.469	5.446	< 0.001	-0.854	0.246
FiLM All	EW	0.868	0.116	5.959	< 0.001	-0.434	0.520
FiLM All	VW	0.463	0.430	5.210	< 0.001	-0.751	0.295
Panel B: 10% ES							
Structured K3	EW	0.889	0.111	6.230	< 0.001	-0.478	0.485
Structured K3	VW	0.609	0.284	6.109	< 0.001	-0.799	0.260
FiLM Sigma	EW	0.913	0.102	6.097	< 0.001	-0.413	0.541
FiLM Sigma	VW	0.414	0.481	5.516	< 0.001	-0.891	0.226
FiLM All	EW	0.842	0.128	5.905	< 0.001	-0.453	0.502
FiLM All	VW	0.420	0.470	5.112	< 0.001	-0.780	0.275

Notes: Portfolios buy the highest-ES stock decile and sell the lowest-ES decile. Both ES levels use the same eligible stocks and separate ES rankings. EW gives stocks equal weights; VW uses formation-month market capitalization. Returns are monthly percentages. Tests use Newey–West standard errors with 12 calendar-month lags. All, bad, and other groups contain 216, 44, and 172 months. Bad market months are the bottom quintile of realized next-month market excess returns; other months are all remaining months. The p-values test zero mean returns and are two-sided. Entries are gross mean return spreads.

mean that the lower-ES stocks have higher average returns in those months, but none of the differences is statistically significant. Full-period equal-weighted means range from 0.84% to 0.91% and are also not significant at 5%. Figure 9 shows both weighting methods and ES levels.

The economic interpretation is that stocks assigned worse ES at t are more vulnerable when $t + 1$ turns out to be a bad market month: they earn significantly lower average realized returns than the highest-ES group. In other months, the average return differences are not significant. This pattern holds for all three structured specifications, both ES levels, and both weighting methods. The ES forecasts therefore identify differences in stocks' vulnerability in bad market months.

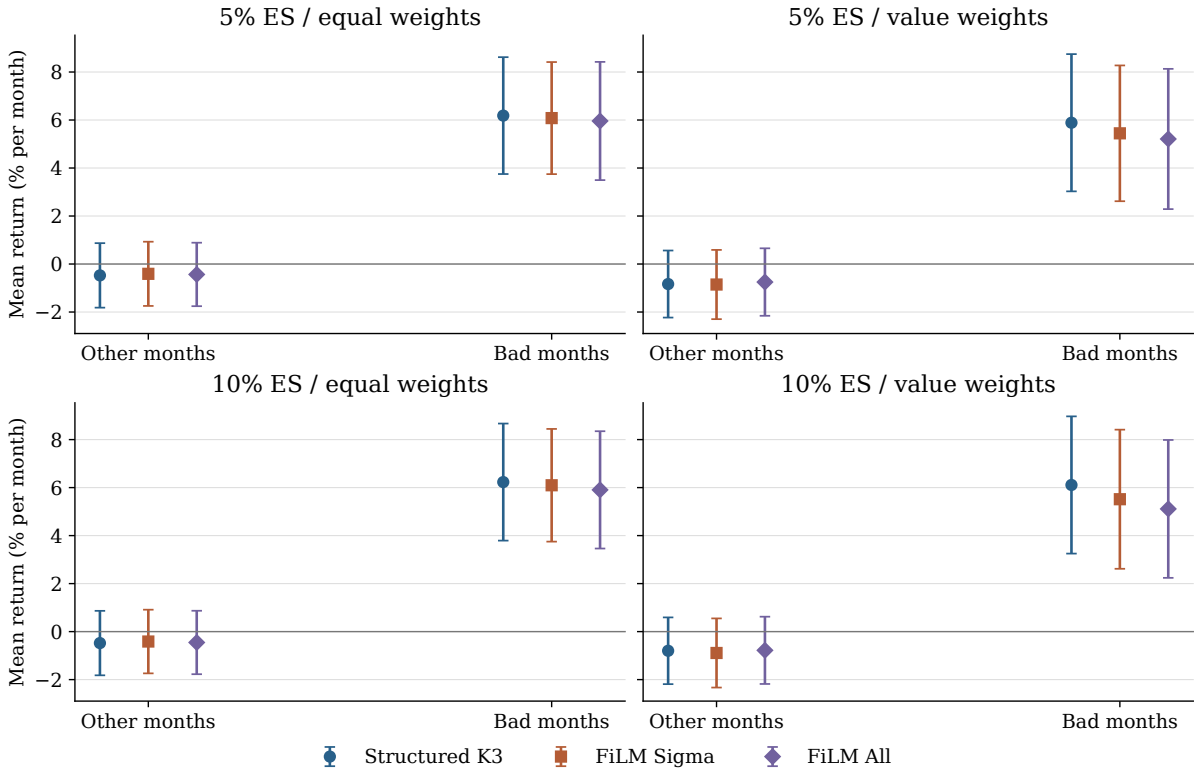


Figure 9: ES-sorted returns in bad market months and other months

Notes: Points are mean highest-minus-lowest ES stock-decile returns in monthly percentages. Rows use 5% and 10% ES; columns use equal and value weights. Both ES levels use the same stocks and the bad market months defined in equation (25). Lines show pointwise 95% intervals based on calendar-time NW12 standard errors. Entries are gross mean return spreads.

4.2 Component Labels and Exact ES Decomposition

At the end of month t, the structured model assigns probabilities $\pi_{1,t}$, $\pi_{2,t}$, and $\pi_{3,t}$ to three possible states in month $t+1$. These state probabilities are identical for every firm. Within state k, however, each firm's return has its own Normal distribution, with mean $\mu_{i,k,t}$ and standard deviation $\sigma_{i,k,t}$. I call this stock-specific distribution component k. Thus, firms face the same state probabilities but can have different returns in the same state.

For each estimation window w, I label the component with the lowest average predicted return across firms over the 84 training and validation months as adverse, denoted A_w. The component with the highest average predicted return is favorable, and the remaining component is middle. These labels apply to every firm and remain fixed throughout the next test year. They are distinct from the bad-market indicator, which classifies realized market returns.

ES is the model's expected return below the stock's VaR cutoff, $q_{i,t,\alpha}$. To decompose it, I first calculate how much of this tail comes from each state, then calculate the average return within each state's part of the tail. All probabilities and expected returns below are conditional on information available at the end of month t.

Within state k, the probability that stock i's return falls below its VaR is $\Phi(z_{i,k,t,\alpha})$, where $z_{i,k,t,\alpha} = (q_{i,t,\alpha} - \mu_{i,k,t})/\sigma_{i,k,t}$ and Φ is the standard Normal distribution function. Multiplying by $\pi_{k,t}$ gives the probability that state k occurs and the stock's return falls below its VaR. Since the total probability below VaR is α, state k's fraction of that tail is

$$\omega_{i,k,t,\alpha} = \frac{\pi_{k,t}\Phi(z_{i,k,t,\alpha})}{\alpha}. \tag{27}$$

This *tail weight* is the probability of state k conditional on the stock's return being below its VaR. The three tail weights sum to one. Although the state probabilities $\pi_{k,t}$ are identical for all firms, tail weights differ across firms because their return distributions and VaR cutoffs differ.

Let $\tau_{i,k,t,\alpha}$ be the expected return in state k conditional on the stock's return being below that same VaR cutoff. The component's contribution is its tail weight multiplied by this conditional expected return. Summing the three contributions gives total ES:

$$C_{i,k,t,\alpha} = \omega_{i,k,t,\alpha}\tau_{i,k,t,\alpha}, \tag{28}$$

$$\mathrm{ES}_{i,t,\alpha} = \sum_{k=1}^{3} C_{i,k,t,\alpha}. \tag{29}$$

I denote the adverse-state contribution to signed ES by $C^A_{i,t,\alpha} = C_{i,A_w,t,\alpha}$ and call it adverse contribution. A more negative value means more predicted loss from the adverse component. Consider $\alpha = 0.05$ and a stock whose VaR is $q_{i,t,0.05} = -20\%$. Suppose the adverse state has probability $\pi_{A_w,t} = 0.30$. Within that state, the probability of the stock's return falling below -20% is $\Phi(z_{i,A_w,t,0.05}) = 0.10$. The adverse state's weight in the stock's overall 5% tail is therefore

$$\omega_{i,A_w,t,0.05} = \frac{0.30 \times 0.10}{0.05} = 0.60.$$

Suppose the stock's expected return in the adverse state, conditional on falling below -20%, is $\tau_{i,A_w,t,0.05} = -0.25$. Equation (28) then gives

$$C^A_{i,t,0.05} = 0.60 \times (-0.25) = -0.15.$$

The adverse component thus contributes -15 percentage points to ES. Adding the middle and favorable components' contributions gives the stock's total ES, as in equation (29).

Two stocks can therefore have the same total ES but different adverse contributions. The following tests examine whether adverse contribution helps distinguish stocks' subsequent losses after accounting for total ES.

Across the three structured models, the component labeled adverse before each test year retains the lowest predicted mean for more than 99.5% of firms in the average test month. Its contribution accounts for 58.4–63.5% of average 5% ES and 55.2–60.1% of average 10% ES. The component contributions also reconstruct total ES to numerical precision. Appendix B.5 provides the detailed labeling rule, Normal tail formulas, and diagnostic table.

4.3 Information in Adverse Contribution Beyond Total ES

Total ES measures a stock's overall predicted tail loss. Adverse contribution measures how much of that loss comes from the adverse component. I test whether this additional detail helps identify stocks that subsequently suffer larger losses, even after accounting for total ES and other risk measures. The question is whether stocks with more negative adverse contributions are more vulnerable in bad market months than their overall ES forecasts alone would suggest.

I use two steps. First, within each month, I relate next-month stock losses to adverse contribution while controlling for total ES and other stock characteristics. This produces one contribution coefficient per month. Second, I compare these coefficients between

bad market months and other months. This shows whether the relation between adverse contribution and subsequent losses is stronger when the market performs poorly. Forecasts and controls are measured at the end of month t; stock returns and the bad-market classification are observed afterward, in month $t+1$.

4.3.1 Stock Outcomes, Sample, and Controls

The main outcome is a stock crash, defined as a loss of more than 10% next month:

$$D_{i,t+1} = \mathbf{1}\{R_{i,t+1} < -0.10\}. \tag{30}$$

The indicator $D_{i,t+1}$ equals one if the stock's return is below -10% and zero otherwise. I also examine the stock's raw return, $R_{i,t+1}$, and the amount it loses beyond the 10% threshold:

$$L_{i,t+1} = \max\{-R_{i,t+1} - 0.10, 0\}. \tag{31}$$

For example, a return of -15% gives $L_{i,t+1} = 0.05$, whereas a return of -8% gives zero. I calculate $L_{i,t+1}$ for every stock, assigning zero whenever the return is at least -10%. I therefore examine three outcomes: whether the stock loses more than 10%, its realized return, and the amount of loss beyond the 10% threshold.

All regressions use the same 410,049 stock–months over 216 months. These observations have the forecasts and historical data required for every control specification. Keeping the sample fixed across models, ES levels, and control sets ensures that differences in results do not arise from using different stocks.

Every contribution regression includes adverse contribution and controls for total ES and industry. I then add other controls in four steps. The first specification includes firm size, the trailing 12-month mean return and volatility, and momentum excluding the formation month. The second adds market beta, which measures sensitivity to market returns. The third adds downside beta and idiosyncratic volatility, motivated by Ang et al. (2006a) and Ang et al. (2006b). Here, downside beta measures sensitivity when market excess returns are below their historical average, and idiosyncratic volatility measures the residual volatility from a market regression. The fourth adds the model's predicted mean and volatility from equations (2) and (3). This final specification tests whether adverse contribution provides additional information about subsequent stock losses after accounting for total ES, historical risk measures, industry, and the model's predicted mean and volatility.

Historical market exposures use the 60 calendar months ending at t, with at least 36 valid stock–market return pairs and 12 pairs for months when market excess returns are

below their average over that historical window. Stocks with insufficient histories are excluded. Appendix A.3 gives the exposure formulas.

4.3.2 Establishing the Relation with Total ES

Before examining adverse contribution, I test whether total ES itself distinguishes stocks' subsequent losses. At the end of each month t, I take each stock's predicted ES and then observe whether the stock crashes in month $t+1$ and how much it loses beyond the 10% threshold. For each model, ES level, and outcome, I estimate a separate regression across stocks within each month:

$$\begin{aligned} Y_{i,t+1} = a_t + b_{E,t}\,\mathrm{ES}^W_{i,t,\alpha} + \boldsymbol{\theta}_t' Z_t^W(X_{i,t}^{(1)}) \\ + \sum_{j \neq j_0} \eta_{j,t} \mathbf{1}\{\text{industry}_{i,t} = j\} + \varepsilon_{i,t+1}. \end{aligned} \tag{32}$$

Here $Y_{i,t+1}$ is either the crash indicator $D_{i,t+1}$ or the loss beyond the 10% threshold $L_{i,t+1}$. The first control set, $X_{i,t}^{(1)}$, includes firm size, historical mean return and volatility, and momentum. The regression also includes an intercept a_t and industry indicators, with j_0 denoting the reference industry. Adverse contribution is not included. The coefficient $b_{E,t}$ measures the relation between total ES and next-month losses after accounting for these controls. All coefficients are estimated separately each month; model and ES-level indices are omitted for readability.

The superscript W indicates that explanatory values are capped at their monthly 1st and 99th percentiles to limit the influence of extremes. ES retains its decimal return units. The transformation Z_t^W additionally standardizes each continuous control to have mean zero and standard deviation one across stocks within the month. Realized outcomes are not capped. These same transformations are used in the contribution regressions below.

Table C.1 averages the monthly ES coefficients over all 216 months (All), the 44 bad market months (Bad), and the 172 other months (Other). The market groups are determined by realized market returns in month $t+1$, after the ES forecasts have been formed. Each reported p-value tests whether the average coefficient in that column is zero, using Newey–West standard errors with 12 calendar-month lags.

Every average coefficient is negative with $p < 0.001$, for both outcomes, all three structured models, and both ES levels, in All, Bad, and Other. Thus, more negative ES forecasts are associated with higher next-month crash probabilities and larger losses beyond the 10% threshold. This relation holds in other months as well as in bad market months, after controlling for historical stock characteristics and industry.

The coefficients are larger in absolute value in bad market months than in other months

for every model, ES level, and outcome. For example, Structured K3 at 5% ES has average crash coefficients of −0.540 over all months, −0.717 in bad months, and −0.495 in other months. A one-percentage-point lower ES, from −0.29 to −0.30, is therefore associated with a 0.717-percentage-point higher crash probability in bad months, compared with 0.495 percentage points in other months, holding the controls fixed. For losses beyond the 10% threshold, the corresponding coefficients are −0.075, −0.150, and −0.056. The same decrease in ES is associated with an additional loss of 0.150 percentage points in bad months, compared with 0.056 percentage points in other months. These estimates indicate a stronger relation between worse predicted ES and subsequent stock losses in bad market months.

Total ES therefore identifies differences in subsequent downside outcomes across stocks under both market conditions, with larger estimated differences in bad months. The contribution analysis asks whether the structured decomposition provides information beyond this already informative overall risk measure.

4.3.3 Step 1: Estimate the Relation Within Each Month

I next add adverse contribution to equation (32) and expand the controls through the four specifications described above. For each structured model and ES level, I estimate a separate regression each month, following the monthly cross-sectional approach of Fama and MacBeth (1973):

$$\begin{aligned} Y_{i,t+1} = a_t^{(s)} + b_{E,t}^{(s)} \, \mathrm{ES}_{i,t,\alpha}^{W} + b_{C,t}^{(s)} C_{i,t,\alpha}^{A,W} + \boldsymbol{\theta}_t^{(s)\prime} Z_t^{W}(X_{i,t}^{(s)}) \\ + \sum_{j \neq j_0} \eta_{j,t}^{(s)} \mathbf{1}\{\text{industry}_{i,t} = j\} + \varepsilon_{i,t+1}^{(s)}. \end{aligned} \tag{33}$$

Here $Y_{i,t+1}$ is the crash indicator, raw return, or loss beyond the 10% threshold. The index s identifies one of the four control sets, and $X_{i,t}^{(s)}$ contains its continuous variables. The regression includes a monthly intercept $a_t^{(s)}$ and industry indicators, with j_0 denoting the reference industry. The model and ES-level indices on the coefficients are omitted for readability.

The key coefficient is $b_{C,t}^{(s)}$. It measures how next-month outcomes differ with adverse contribution after accounting for total ES and the controls. For crash probability, a negative coefficient means that a more negative adverse contribution predicts a higher probability of a stock crash. For example, if $b_{C,t}^{(s)} = -2$, a contribution of −0.21 rather than −0.20 is associated with a two-percentage-point higher crash probability, holding the other regressors fixed. Because total ES is included, this comparison asks whether the allocation of predicted tail losses across components matters beyond their total.

Adverse contribution is capped at its own monthly 1st and 99th percentiles and retains its decimal return units. The exact ES decomposition holds for the original forecasts, although separately capping ES and contribution need not preserve that identity.

4.3.4 Step 2: Compare the Relation Across Market Conditions

The first step produces 216 monthly estimates of the contribution coefficient. I then test whether these coefficients differ between the 44 bad market months and the 172 other months defined in equation (25):

$$\widehat{b}_{C,t}^{(s)} = \kappa^{(s)} + \delta_C^{(s)} B_{t+1} + \sum_{w=1}^{17} \lambda_w^{(s)} \mathbf{1}\{w(t) = w\} + u_t^{(s)}. \tag{34}$$

The dependent variable is the estimated monthly coefficient $\widehat{b}_{C,t}^{(s)}$, rather than an individual stock's return or crash indicator. The bad-market indicator B_{t+1} equals one in bad market months and zero in other months. The coefficient $\delta_C^{(s)}$ measures how much the contribution coefficient changes in bad market months.

For crashes, a negative $\delta_C^{(s)}$ indicates that, after controlling for total ES and the other stock characteristics, a more negative adverse contribution is more strongly associated with higher crash probability in bad market months than in other months.

The annual-window indicators $\mathbf{1}\{w(t) = w\}$ allow the average contribution coefficient to differ across the 18 annual estimation windows; window 0 is the reference. This accounts for changes in average coefficients when the forecasting model is re-estimated each year. Standard errors use the Newey–West method with 12 calendar-month lags to account for serial correlation across monthly coefficients. Appendix B.7 gives the calculation details.

4.3.5 Results for Stock Crashes

Table 9 reports δ_C for all three structured models, both ES levels, and the four control sets. Each entry therefore tests whether adverse contribution is more strongly related to crash probability in bad market months, after accounting for total ES and the listed controls.

All 24 estimates are negative and statistically significant at 5%. The result survives the addition of market beta, downside beta, idiosyncratic volatility, and the model's predicted mean and volatility. It therefore cannot be explained solely by the linear relations with total ES and these measured risk characteristics.

With all controls included, δ_C is -4.114, -2.386, and -2.343 at 5% ES for Structured K3, FiLM Sigma, and FiLM All, respectively. At 10% ES, the corresponding estimates are -6.239, -3.767, and -3.495. All six have $p < 0.001$. In each specification, a more

Table 9: Tail contribution and bad-market crash sensitivity at both ES levels

Model	Controls	δ_C: 5%	SE	p	δ_C: 10%	SE	p
Structured K3	1 History	-4.240	1.212	< 0.001	-6.319	1.686	< 0.001
Structured K3	2 + market beta	-3.542	1.117	0.002	-5.290	1.563	< 0.001
Structured K3	3 + downside / idiosyncratic volatility	-3.573	1.119	0.001	-5.342	1.563	< 0.001
Structured K3	4 + predicted mean / volatility	-4.114	1.227	< 0.001	-6.239	1.883	< 0.001
FiLM Sigma	1 History	-2.454	0.350	< 0.001	-3.644	0.531	< 0.001
FiLM Sigma	2 + market beta	-2.005	0.315	< 0.001	-3.002	0.482	< 0.001
FiLM Sigma	3 + downside / idiosyncratic volatility	-2.018	0.299	< 0.001	-3.028	0.457	< 0.001
FiLM Sigma	4 + predicted mean / volatility	-2.386	0.345	< 0.001	-3.767	0.544	< 0.001
FiLM All	1 History	-2.314	0.381	< 0.001	-3.398	0.583	< 0.001
FiLM All	2 + market beta	-1.869	0.339	< 0.001	-2.744	0.513	< 0.001
FiLM All	3 + downside / idiosyncratic volatility	-1.892	0.326	< 0.001	-2.785	0.496	< 0.001
FiLM All	4 + predicted mean / volatility	-2.343	0.363	< 0.001	-3.495	0.580	< 0.001

Notes: All specifications use the same 410,049 stock–months over 216 months. Each monthly stock regression includes total ES, adverse contribution, industry indicators, and the listed controls. History includes log size, historical mean return and volatility, and momentum. The second regression compares monthly contribution coefficients between bad market months and other months, controlling for annual estimation-window indicators. Each entry is this adjusted difference, δ_C, in crash-probability percentage points per one-percentage-point increase in contribution. Negative values indicate a stronger association between more negative contribution and higher crash probability in bad market months. Two-sided p-values use Newey–West standard errors with 12 calendar-month lags.

negative adverse contribution is associated with a larger increase in crash probability in bad months than in other months, holding total ES and the controls fixed.

For Structured K3 at 5% ES, $\delta_C = -4.114$. Holding total ES and the controls fixed, a decrease in adverse contribution from -0.20 to -0.21 is associated with a larger increase in crash probability in bad market months than in other months. After accounting for annual-window effects, the difference is 4.114 percentage points.

4.3.6 Results for Returns and Loss Amounts

The crash indicator records whether a stock loses more than 10%, but does not measure its return or the amount lost beyond that threshold. I therefore repeat the two-step analysis using these two outcomes, controlling for total ES, historical stock characteristics, industry, and the model's predicted mean and volatility.

For raw returns, all six estimates of δ_C are positive and significant at 1%. For losses beyond the 10% threshold, all six are negative and significant at 5%. Both results indicate worse outcomes for stocks with more negative adverse contributions in bad market months relative to other months. Table C.2 reports the estimates.

For example, Structured K3 at 5% ES has $\delta_C = 0.853$ for returns and $\delta_C = -0.317$ for losses beyond the threshold. Holding total ES and the controls fixed, a one-percentage-point

lower adverse contribution is associated with a return difference that is 0.853 percentage points more negative in bad months than in other months, and a difference in losses beyond the 10% threshold that is 0.317 percentage points larger.

Together with the crash results, these findings show that adverse contribution provides information about both the likelihood and the extent of subsequent stock losses. The evidence holds across all three structured models and both ES levels, supporting the economic value of decomposing total ES rather than using the overall forecast alone.

4.4 Contribution Residuals and Crash Rates

The preceding regressions show that adverse contribution helps identify stocks with larger subsequent losses after accounting for total ES and other risk measures. I now compare the next-month crash rates of two stock groups formed using adverse contribution, adjusted for those same risk measures. This gives the difference in the fraction of stocks that actually lose more than 10%, expressed in percentage points.

4.4.1 Adjusting Contribution and Forming Stock Groups

Stocks with more negative adverse contributions may also have more negative total ES or higher volatility. If these stocks crash more often, the difference could reflect their overall tail risk or volatility. To account for these characteristics before forming stock groups, I first regress adverse contribution on total ES and the full control set across stocks within each formation month:

$$C^{A,W}_{i,t,\alpha} = a_t + \rho_t \, \mathrm{ES}^{W}_{i,t,\alpha} + \boldsymbol{\psi}_t' Z^{W}_t(X^{\mathrm{full}}_{i,t}) + \sum_{j \neq j_0} \xi_{j,t} \mathbf{1}\{\mathrm{industry}_{i,t} = j\} + \widetilde{C}^{A}_{i,t,\alpha}. \tag{35}$$

Here $X^{\mathrm{full}}_{i,t}$ includes firm size, historical mean return and volatility, momentum, market beta, downside beta, idiosyncratic volatility, and the model's predicted mean and volatility. Industry indicators are also included, with j_0 denoting the reference industry. The superscript W and transformation Z^W_t follow the same capping and standardization rules as the preceding regressions. The analysis uses the same 410,049 stock–months and is conducted separately for each structured model and ES level.

For each stock, this regression gives a fitted adverse contribution based on its total ES, controls, and industry. I subtract that fitted value from the stock's adverse contribution entering the regression to obtain the residual $\widetilde{C}^{A}_{i,t,\alpha}$. For example, if the stock's adverse contribution is -0.24 and the fitted value is -0.20, the residual is -0.04. The stock's adverse contribution is therefore four percentage points more negative than the value fitted

using its total ES and other characteristics.

At the end of month t, I rank stocks by these residuals and select the bottom 10% and top 10%. The bottom group has the most negative differences between adverse contribution and its fitted value; the top group has the largest differences. I calculate the residuals and form both groups using information available at t, before observing next-month returns.

4.4.2 Calculating and Comparing Crash Rates

After month $t+1$ ends, I calculate each group's crash rate as the fraction of its stocks with realized returns below -10%. The monthly crash-rate gap is the bottom group's rate minus the top group's rate:

$$G_{t+1} = \frac{1}{N_{1,t}} \sum_{i \in \mathcal{D}_{1,t}} D_{i,t+1} - \frac{1}{N_{10,t}} \sum_{i \in \mathcal{D}_{10,t}} D_{i,t+1}. \tag{36}$$

Here $\mathcal{D}_{1,t}$ and $\mathcal{D}_{10,t}$ are the bottom and top residual groups, containing $N_{1,t}$ and $N_{10,t}$ stocks. The crash indicator $D_{i,t+1}$ is defined in equation (30). Model and ES-level indices are omitted for readability. A positive gap means that a larger fraction of stocks crash in the bottom group. I report gaps in percentage points, using $100G_{t+1}$.

Table 10 reports the average monthly gap over the 44 bad market months (Bad gap) and the 172 other months (Other gap). These market groups are determined using realized market returns after the stock groups have been formed. The p-values test whether the average gap is zero within bad and other months, respectively. Standard errors use Newey–West with 12 calendar-month lags, preserving the actual spacing between selected months.

To test whether the crash-rate gap is larger in bad market months, I regress the 216 monthly gaps, expressed in percentage points, on the bad-market indicator and annual estimation-window indicators:

$$100G_{t+1} = \kappa_G + \Delta B_{t+1} + \sum_{w=1}^{17} \lambda_{G,w} \mathbf{1}\{w(t) = w\} + u_{G,t+1}. \tag{37}$$

Here B_{t+1} equals one in bad market months and zero in other months. The index $w(t)$ identifies the annual estimation window used to form the stock groups. The window indicators allow the average gap to differ across the 18 estimation windows, with window 0 as the reference.

The coefficient Δ is reported as Adjusted Δ in Table 10. A positive value means that the bottom-minus-top crash-rate gap is larger in bad market months after accounting for these annual differences. For example, $\Delta = 6.710$ for Structured K3 at 5% ES means

that the gap is 6.710 percentage points larger in bad months than in other months. The reported p-value tests $\Delta = 0$ using a two-sided test with Newey–West standard errors and 12 calendar-month lags.

4.4.3 Results and Economic Interpretation

Table 10: Sorts on contribution residuals at the 5% and 10% ES levels

Model	ES (%)	Bad gap	p	Other gap	p	Adjusted Δ	p
Structured K3	5	6.449	< 0.001	0.091	0.836	6.710	< 0.001
Structured K3	10	6.463	< 0.001	-0.003	0.994	6.757	< 0.001
FiLM Sigma	5	7.141	< 0.001	-0.406	0.294	7.944	< 0.001
FiLM Sigma	10	7.596	< 0.001	-0.205	0.624	8.202	< 0.001
FiLM All	5	6.588	< 0.001	0.093	0.806	6.817	< 0.001
FiLM All	10	6.207	< 0.001	0.162	0.656	6.343	< 0.001

Notes: The sample contains 410,049 stock–months over 216 months. Stocks are ranked each month by adverse contribution minus its fitted value from a regression on total ES, the full control set, and industry indicators. The gap is the bottom 10% group's next-month crash rate minus the top 10% group's rate, in percentage points. A crash is a realized stock return below -10%. Bad gap and Other gap are average monthly gaps over 44 bad and 172 other months; their p-values test zero average gaps. Adjusted Δ is the coefficient on the bad-market indicator in a regression of monthly gaps on that indicator and annual estimation-window indicators; its p-value tests a zero coefficient. All p-values are two-sided and use Newey–West standard errors with 12 calendar-month lags, preserving the calendar spacing of selected months.

In bad market months, the bottom residual group has a substantially higher crash rate than the top group. At 5% ES, the gaps are 6.45 percentage points for Structured K3, 7.14 for FiLM Sigma, and 6.59 for FiLM All. At 10% ES, they are 6.46, 7.60, and 6.21 percentage points, respectively. All six average gaps have $p < 0.001$. For example, the 6.45-percentage-point gap for Structured K3 means that, in the average bad month, the fraction of stocks losing more than 10% is 6.45 percentage points higher in the bottom group than in the top group.

In other months, the average gaps range from -0.41 to 0.16 percentage points; none is statistically different from zero. The adjusted bad-minus-other differences range from 6.34 to 8.20 percentage points, all with $p < 0.001$. Thus, the stock groups formed in advance differ clearly in their subsequent crash rates in bad market months, while the tests find no significant average gap in other months. This pattern holds for all three structured models and both ES levels.

Figure 10 displays the Bad gap and Other gap estimates with 95% confidence intervals. Every bad-month interval is above zero, whereas every other-month interval includes zero.

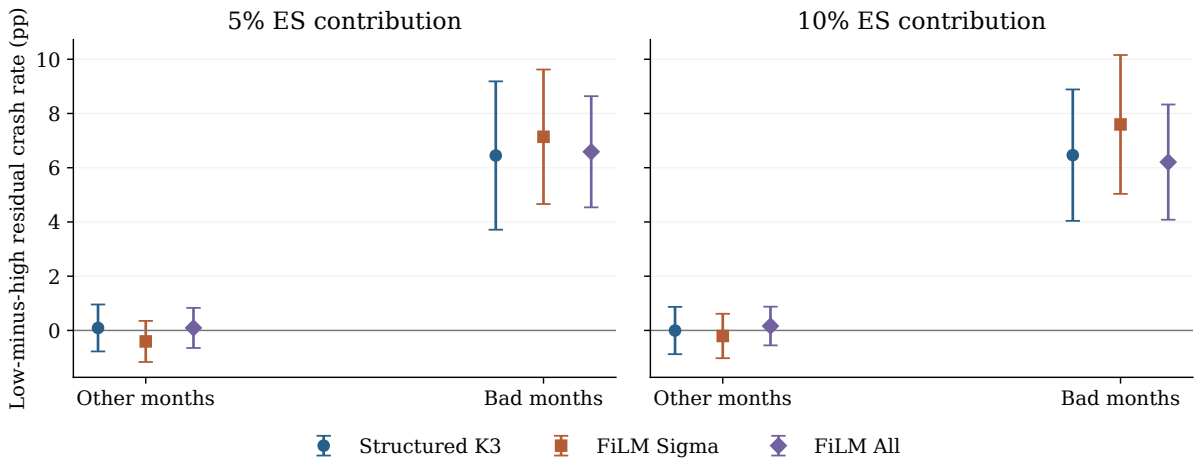


Figure 10: Crash-rate differences between low and high contribution-residual groups

Notes: At the end of each month, I regress adverse contribution on total ES, the full control set, and industry indicators, and rank stocks by contribution minus its fitted value. The bottom and top 10% form the two stock groups. Each point shows the bottom group's next-month crash rate minus the top group's rate, averaged over the 44 bad or 172 other months. A crash is a realized stock return below −10%. The left panel uses 5% ES contributions and the right panel uses 10% ES contributions. Bad market months are defined in equation (25). Bars are pointwise 95% confidence intervals based on Newey–West standard errors with 12 calendar-month lags. Units are percentage points.

Table C.4 compares the two groups' risk characteristics. In bad market months, the bottom group's average total ES minus the top group's ranges from −0.35 to 0.36 percentage points across the three models and two ES levels. A negative difference means worse average ES in the bottom group; a positive difference means better average ES. The two groups therefore have similar average total ES, and the bottom group does not consistently have the worse forecast. Its average market beta is also slightly lower in all six comparisons. Thus, the bottom group's higher crash rate occurs even though it does not consistently have worse average total ES or higher average sensitivity to market returns.

Total ES and the controls explain most of the differences in adverse contribution across stocks. Table C.3 measures how much remains by dividing the variance of the contribution residuals by the variance of winsorized adverse contribution before the regression adjustment. I calculate this ratio within each month and then average across months. The residuals retain 5.8%–9.4% of this contribution variance across models and ES levels. These residuals identify stocks whose adverse contributions are lower or higher than the values fitted using total ES and the other controls.

Although this remaining variation is a small share of total contribution variance, it distinguishes stocks with substantially different subsequent crash rates. The bottom residual group has a crash rate about 6–8 percentage points higher than the top group in bad market months. These groups are selected before next-month returns are observed.

Adverse contribution therefore provides additional information about which stocks are more likely to crash in bad market months, after accounting for total ES and the other included risk measures.

4.5 Economic Information in the Structured Forecasts

Total ES identifies stocks that subsequently suffer greater losses. Groups formed at month-end from more negative ES forecasts have lower returns in bad market months under both weighting methods. Monthly regressions also link more negative ES to higher crash probabilities and larger losses beyond 10%, after controlling for historical stock characteristics and industry.

Adverse contribution provides additional information. Its relation with crashes, returns, and loss amounts becomes stronger in bad market months even after controlling for total ES, historical risk measures, industry, and predicted mean and volatility. Groups formed from the bottom and top contribution residuals differ in bad-month crash rates by about 6–8 percentage points across all three models and both ES levels. Their average total ES is similar, and the bottom group has slightly lower average market beta.

Two stocks can have similar total ES even when the adverse state contributes different amounts to their predicted tail losses. The residual-group results show that these differences help identify stocks that crash more often in bad market months. This is the economic value of comparing state contributions alongside total ES and the included risk measures.

5 Robustness and Training Sensitivity

I check whether the contribution results persist under different bad-market definitions and Newey–West lag lengths. I also assess both forecast accuracy and the economic results under different random seeds and a fixed training duration. The additional training runs use the same predictors, including formation-month return, and are evaluated over the same 216 months.

5.1 Bad-Market Definitions and Standard Errors

The main definition selects the 20% of months with the lowest realized market excess returns. I repeat the contribution test using fixed cutoffs of -3% and -5%, which identify 40 and 25 bad months, respectively. For both cutoffs, all three structured models, and both ES levels, the fully controlled contribution coefficient is more negative in bad months than in other months, with every difference significant at 5%. Thus, more negative adverse

contribution remains associated with a greater increase in crash probability in bad months when those months are defined by a fixed market-loss threshold.

I also recalculate the standard errors using 3, 6, and 12 calendar-month Newey–West lags under the main bad-market definition. This changes the allowance for serial correlation while leaving the coefficient estimates unchanged. All six fully controlled contribution slope differences remain negative with $p < 0.001$ at every lag length. Tables D.1 and D.2 in Appendix D report these checks. The contribution signal therefore identifies greater crash vulnerability under both percentile-based and fixed-loss definitions of bad market months. Its statistical significance also survives different allowances for serial correlation. These checks support the economic finding across reasonable choices of market definition and standard-error calculation.

5.2 Random Seeds

To assess sensitivity to randomness in training, I repeat the validation-based procedure with seeds 2026 and 42 alongside the main seed 2027. For each seed, I train all five neural models. This gives nine structured model–seed combinations, each compared with fused benchmarks trained using the same seed. In every combination, the structured model has lower NLL and CRPS, as well as lower FZ0 at both ES levels than Fused K1 and Fused K3 trained under the same configuration (Table E.5).

The training records show the same pattern across all three validation seeds. None of the structured models selects epoch 1, and their median selected epochs range from 20 to 30, compared with 1 to 4 for the fused models. For every seed, each structured model's median absolute change in validation NLL between consecutive epochs is less than half that of either fused benchmark, using the measure defined in Section 3.5. Thus, the continued training and smaller changes between epochs observed in the main run also occur under the other initializations (Table E.1).

The economic results also recur. At both ES levels, every fully controlled bad-minus-other contribution slope difference is negative with $p < 0.001$. The bottom-minus-top residual-group crash-rate gaps in bad months range from 3.58 to 7.60 percentage points, all significant at 1%. In other months, no gap is significant at 5% ES. At 10% ES, one gap is significant and negative: FiLM Sigma with seed 2026 has a gap of -0.84 percentage points ($p = 0.039$). The other eight are not significant. Tables E.2 and E.3 report each model and seed separately.

The three structured variants also continue to select similar stocks. In every run, at both ES levels and in both extreme ES deciles, overlap between any two structured models

exceeds overlap between any structured model and either fused benchmark (Table E.6). The best-performing structured variant can change with the seed and forecast metric, but all three retain their advantage over the fused benchmarks and provide useful contribution signals. The results support the forecast advantage and useful contribution information across different training initializations.

5.3 Fixed Training Duration

Under validation-based selection, the structured models generally retain later epochs than the fused models. I therefore check whether their forecast advantage remains when both families train for the same number of epochs. Using seed 2027, I train all five neural models for exactly 40 epochs in each annual window and retain epoch 40. Each model uses the same 84 months formed by combining the 72 training months and 12 validation months, with a constant learning rate and no validation-based checkpoint selection. Equal epoch counts do not imply equal numbers of parameter updates, because the structured and fused procedures use different batch constructions.

All three structured models still have lower NLL and CRPS, as well as lower FZ0 at both ES levels, than both fused benchmarks (Table E.5). Their CRPS values range from 0.0717 to 0.0735, compared with 0.0778–0.0796 for the fused models. The forecast advantage therefore persists under a fixed-epoch protocol without validation-based checkpoint selection. This result supports the structured architecture and its common-state training procedure as an effective way to combine macroeconomic and firm information.

The economic information in adverse contribution also remains useful under fixed-duration training. All six fully controlled contribution slope differences are negative with $p < 0.001$. Bad-month residual crash-rate gaps range from 5.68 to 7.03 percentage points across models and ES levels, all with $p < 0.001$. The adjusted bad-minus-other gaps are also positive with $p < 0.001$ in all six comparisons. FiLM All has a smaller positive gap of 0.71 percentage points in other months at 5% ES ($p = 0.038$), compared with 7.03 percentage points in bad months; the other five other-month gaps are not significant. Thus, the ability to identify stocks with greater bad-market crash vulnerability persists when training ends at a predetermined epoch, without selecting a checkpoint using validation outcomes.

Appendix E reports all four training runs, together with the mixture–Normal comparisons and ES portfolio results. Across these checks, the structured models produce more accurate forecasts and their adverse contributions identify differences in subsequent stock losses beyond total ES and the other controls. The results support both parts

of the proposed approach: structuring the return distribution around common states improves forecasting, and decomposing its tail losses provides useful information about firms' downside vulnerability.

6 Conclusion

This paper proposes a structured MDN in which macroeconomic inputs determine common state probabilities and each firm has its own return distribution within each state. The three specifications differ in whether macroeconomic information also adjusts the learned firm features used to predict component means and standard deviations. Joint training uses complete monthly stock cross-sections to estimate the common probabilities and firm-specific distributions. This design gives the adverse component the same label across firms and allows its contribution to each stock's ES to be compared consistently.

The three structured specifications improve on both fused neural benchmarks across all eight forecast scores. The mixtures also improve density forecasts relative to Normal distributions with identical predicted means and volatilities. These advantages recur across training seeds and remain when all neural models train for 40 epochs and use their final checkpoints. The forecasting evidence supports the structured design and its common-state training procedure.

The forecasts provide useful information about subsequent stock losses. Total ES identifies stocks that perform worse in bad market months. Adverse contribution adds information after accounting for total ES, historical risk measures, industry, and predicted mean and volatility. Groups formed in advance from contribution residuals have bad-month crash rates that differ by about 6–8 percentage points in the main analysis. The positive crash-rate gaps recur across the additional training runs. Thus, the structured model supplies both an overall tail-risk forecast and a component contribution that helps identify further differences in downside vulnerability.

The broader contribution is a structured approach to forecasting return distributions that also makes the sources of predicted tail losses comparable across firms. The common states give each component contribution a consistent meaning, and the empirical results show why this matters: adverse-state contributions distinguish subsequent stock losses in bad market months after controlling linearly for total ES and the included risk measures. The framework thus improves distribution forecasts and provides economically useful information about stock losses beyond the overall severity of predicted tail risk.

References

Acerbi, C. and Tasche, D. (2002). Expected shortfall: A natural coherent alternative to value at risk. *Economic Notes*, 31(2):379–388.

Ang, A. and Bekaert, G. (2002). International asset allocation with regime shifts. *Review of Financial Studies*, 15(4):1137–1187.

Ang, A. and Chen, J. (2002). Asymmetric correlations of equity portfolios. *Journal of Financial Economics*, 63(3):443–494.

Ang, A., Chen, J., and Xing, Y. (2006a). Downside risk. *Review of Financial Studies*, 19(4):1191–1239.

Ang, A., Hodrick, R. J., Xing, Y., and Zhang, X. (2006b). The cross-section of volatility and expected returns. *Journal of Finance*, 61(1):259–299.

Artzner, P., Delbaen, F., Eber, J.-M., and Heath, D. (1999). Coherent measures of risk. *Mathematical Finance*, 9(3):203–228.

Atilgan, Y., Bali, T. G., Demirtas, K. O., and Gunaydin, A. D. (2020). Left-tail momentum: Underreaction to bad news, costly arbitrage and equity returns. *Journal of Financial Economics*, 135(3):725–753.

Baruník, J., Hronec, M., and Tobek, O. (2026). Forecasting stock return distributions around the globe with quantile neural networks. *International Journal of Forecasting*, 42(4):1298–1313.

Bishop, C. M. (1994). Mixture density networks. Technical Report NCRG/94/004, Aston University.

Chen, J., Hong, H., and Stein, J. C. (2001). Forecasting crashes: Trading volume, past returns, and conditional skewness in stock prices. *Journal of Financial Economics*, 61(3):345–381.

Christoffersen, P. F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4):841–862.

Denault, M. (2001). Coherent allocation of risk capital. *Journal of Risk*, 4(1):1–34.

Diebold, F. X., Gunther, T. A., and Tay, A. S. (1998). Evaluating density forecasts with applications to financial risk management. *International Economic Review*, 39(4):863–883.

Diebold, F. X. and Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263.

Diks, C., Panchenko, V., and van Dijk, D. (2011). Likelihood-based scoring rules for comparing density forecasts in tails. *Journal of Econometrics*, 163(2):215–230.

Dittmar, R. F. (2002). Nonlinear pricing kernels, kurtosis preference, and evidence from the cross section of equity returns. *Journal of Finance*, 57(1):369–403.

Engle, R. F. and Manganelli, S. (2004). CAViaR: Conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, 22(4):367–381.

Fama, E. F. and MacBeth, J. D. (1973). Risk, return, and equilibrium: Empirical tests. *Journal of Political Economy*, 81(3):607–636.

Fissler, T. and Ziegel, J. F. (2016). Higher order elicitability and osband's principle. *Annals of Statistics*, 44(4):1680–1707.

Giacomini, R., Gottschling, A., Haefke, C., and White, H. (2008). Mixtures of t-distributions for finance and forecasting. *Journal of Econometrics*, 144(1):175–192.

Giacomini, R. and White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6):1545–1578.

Gneiting, T. (2011). Making and evaluating point forecasts. *Journal of the American Statistical Association*, 106(494):746–762.

Gneiting, T., Balabdaoui, F., and Raftery, A. E. (2007). Probabilistic forecasts, calibration and sharpness. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 69(2):243–268.

Gneiting, T. and Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378.

Gneiting, T. and Ranjan, R. (2011). Comparing density forecasts using threshold- and quantile-weighted scoring rules. *Journal of Business & Economic Statistics*, 29(3):411–422.

Gray, S. F. (1996). Modeling the conditional distribution of interest rates as a regime-switching process. *Journal of Financial Economics*, 42(1):27–62.

Gu, S., Kelly, B., and Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5):2223–2273.

Gu, S., Kelly, B., and Xiu, D. (2021). Autoencoder asset pricing models. *Journal of Econometrics*, 222(1):429–450.

Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384.

Hansen, B. E. (1994). Autoregressive conditional density estimation. *International Economic Review*, 35(3):705–730.

Harvey, C. R. and Siddique, A. (2000). Conditional skewness in asset pricing tests. *Journal of Finance*, 55(3):1263–1295.

Huang, W., Liu, Q., Rhee, S. G., and Wu, F. (2012). Extreme downside risk and expected stock returns. *Journal of Banking & Finance*, 36(5):1492–1502.

Jegadeesh, N. (1990). Evidence of predictable behavior of security returns. *Journal of Finance*, 45(3):881–898.

Jegadeesh, N. and Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1):65–91.

Jordan, M. I. and Jacobs, R. A. (1994). Hierarchical mixtures of experts and the EM algorithm. *Neural Computation*, 6(2):181–214.

Kelly, B. and Jiang, H. (2014). Tail risk and asset prices. *Review of Financial Studies*, 27(10):2841–2871.

Kelly, B. T., Pruitt, S., and Su, Y. (2019). Characteristics are covariances: A unified model of risk and return. *Journal of Financial Economics*, 134(3):501–524.

Koenker, R. and Bassett, Jr., G. (1978). Regression quantiles. *Econometrica*, 46(1):33–50.

Loshchilov, I. and Hutter, F. (2019). Decoupled weight decay regularization. In *International Conference on Learning Representations*.

McCracken, M. W. and Ng, S. (2016). FRED-MD: A monthly database for macroeconomic research. *Journal of Business & Economic Statistics*, 34(4):574–589.

McNeil, A. J. and Frey, R. (2000). Estimation of tail-related risk measures for heteroscedastic financial time series: An extreme value approach. *Journal of Empirical Finance*, 7(3–4):271–300.

Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708.

Patton, A. J., Ziegel, J. F., and Chen, R. (2019). Dynamic semiparametric models for expected shortfall (and value-at-risk). *Journal of Econometrics*, 211(2):388–413.

Perez, E., Strub, F., de Vries, H., Dumoulin, V., and Courville, A. (2018). FiLM: Visual reasoning with a general conditioning layer. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32.

Petersen, M. A. (2009). Estimating standard errors in finance panel data sets: Comparing approaches. *Review of Financial Studies*, 22(1):435–480.

Rockafellar, R. T. and Uryasev, S. (2000). Optimization of conditional value-at-risk. *Journal of Risk*, 2(3):21–41.

Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., and Salakhutdinov, R. (2014). Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(56):1929–1958.

Stephens, M. (2000). Dealing with label switching in mixture models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 62(4):795–809.

Taylor, J. W. (2019). Forecasting value at risk and expected shortfall using a semiparametric approach based on the asymmetric laplace distribution. *Journal of Business & Economic Statistics*, 37(1):121–133.

A Data, Predictors, and Regression Controls

A.1 Firm Data

The models use 21 continuous firm inputs: 15 accounting ratios, five derived market variables, and the raw formation-month return, `mthret`. The last variable is observed in month t and is used to forecast the return in $t + 1$. Price is used in sample eligibility and variable construction but is not a separate continuous neural input. I first build the calendar grid and next-month target, then restrict formation months to 2000–2024. Only formation months in 2007–2024 are used for out-of-sample evaluation.

Table A.1: Continuous firm predictors

Field	Description
`bm`	Book-to-market
`evm`	Enterprise-value multiple
`ps`	Price-to-sales
`pcf`	Price-to-cash-flow
`opmad`	Operating profit margin after depreciation
`cfm`	Cash-flow margin
`roe`	Return on equity
`gprof`	Gross profitability
`debt_at`	Debt-to-assets
`short_debt`	Short-term debt share
`cash_ratio`	Cash ratio
`curr_ratio`	Current ratio
`at_turn`	Asset turnover
`rd_sale`	Research and development to sales
`accrual`	Accruals
`MOM12m`	Compounded returns over months $t-11$ through $t-1$; at least six valid months
`mu_bench`	Mean return over months $t-11$ through t; at least six valid months
`sigma_bench`	Sample return standard deviation over months $t-11$ through t; at least six valid months
`turnover`	Monthly volume divided by shares outstanding (in shares)
`mthcap_log`	Log formation-month market capitalization
`mthret`	Raw return in formation month t; predicts the return in month $t+1$

Notes: Accounting ratios follow the WRDS definitions. Market variables use the complete CRSP calendar panel. Turnover divides monthly volume by stored shares outstanding multiplied by 1,000 to obtain shares. Industry enters separately as a learned embedding, with a category for missing classifications.

Accounting ratios are linked to CRSP firms through the Compustat link and recorded public date, then carried forward to the formation month. The date check confirms that the recorded public date is no later than formation. It does not reconstruct revisions that are absent from the stored data. The five derived predictors are calculated before price and industry filters, so removing a stock from the sample does not compress its calendar history.

A.2 Macro Data

Table A.2: Macroeconomic and financial information

Series	Source	Description
PAYEMS	FRED-MD	Nonfarm employment; vintage transformation
UNRATE	FRED-MD	Unemployment rate
INDPRO	FRED-MD	Industrial production; vintage transformation
RPI	FRED-MD	Real personal income; vintage transformation
ISRATIOx	FRED-MD	Inventories-to-sales ratio; vintage transformation
HOUST	FRED-MD	Housing starts; vintage transformation
BUSLOANS	FRED-MD	Business loans; vintage transformation
CPIAUCSL	FRED-MD	Consumer price index; vintage transformation
VIXCLS	FRED	Monthly average VIX
TERM_SPREAD	FRED	10-year Treasury yield minus 3-month bill yield
CREDIT_SPREAD	FRED	Baa corporate yield minus 10-year Treasury yield
FEDFUNDS	FRED	Monthly change in averaged effective policy rate
sprtrn	WRDS	S&P 500 monthly return

Notes: FRED-MD transformations are applied within each forecast-month vintage. The eight FRED-MD series use their specified transformations where required. Other inputs are measured monthly. All 13 variables are scaled using one observation per training month.

Each forecast month has a corresponding FRED-MD vintage file. For 2000–2014, I use the historical vintages reconstructed by the St. Louis Fed from archived Haver Analytics data, retaining the latest version available by the corresponding month-end. From 2015 onward, I use the published monthly vintages. I apply the supplied transformation code within each file and select the latest nonmissing transformed value for each series. The observation month can therefore differ across series in the same vintage. Additional financial series use monthly averages of daily FRED values. The policy-rate variable is the monthly change in the averaged effective federal funds rate, and the equity-market variable is the S&P 500 monthly return. Missing values are filled with medians from the training months. Means and standard deviations for scaling also use one observation per

training month. Validation and test data use these same values.

A.3 Historical Market and Downside Exposures

This subsection defines the historical exposure controls used in Section 4.3.1. They are calculated from returns dated no later than formation month t. The estimates measure past market sensitivity and idiosyncratic volatility. The MDN's predicted mean and standard deviation describe the future return distribution.

Let $\mathcal{H}_t = \{t-59, \ldots, t\}$ be the 60 calendar months ending at formation. Within $\mathcal{H}_t$, let $\mathcal{P}_{i,t}$ contain the months with valid stock and market excess-return pairs. Here s indexes a historical month, $R_{i,s}$ is the raw stock return, and $R_{f,s}$ is that month's risk-free return. The stock excess return is $r^e_{i,s} = R_{i,s} - R_{f,s}$, and M^e_s is the market excess return. Estimate the following market regression with an intercept:

$$r^e_{i,s} = a_{i,t} + \beta_{i,t} M^e_s + u^{(t)}_{i,s}, \qquad s \in \mathcal{P}_{i,t}. \tag{38}$$

The intercept is $a_{i,t}$, the historical market beta is $\beta_{i,t}$, and $u^{(t)}_{i,s}$ is the historical regression disturbance for the regression estimated at formation month t. The exposure is indexed by t because it is re-estimated using the history available at each formation month. The regression requires at least 36 valid pairs and positive market-return variation. Idiosyncratic volatility is the standard deviation of the market-model residuals, calculated as

$$\sigma^{\text{idio}}_{i,t} = \left[\frac{\sum_{s \in \mathcal{P}_{i,t}} (\widehat{u}^{(t)}_{i,s})^2}{|\mathcal{P}_{i,t}| - 2} \right]^{1/2}. \tag{39}$$

Here $\widehat{u}^{(t)}_{i,s}$ is the fitted residual for the regression estimated at formation month t, $\sigma^{\text{idio}}_{i,t}$ is idiosyncratic volatility, and the label idio means "idiosyncratic." The denominator subtracts two degrees of freedom for the estimated intercept and market slope. Define the downside subset $\mathcal{P}^-_{i,t}$ as

$$\mathcal{P}^-_{i,t} = \{s \in \mathcal{P}_{i,t} : M^e_s < \overline{M}^e_{\mathcal{H}_t}\}. \tag{40}$$

The superscript minus labels the downside subset. The threshold $\overline{M}^e_{\mathcal{H}_t}$ is the average market excess return over the full calendar window $\mathcal{H}_t$, including months when the stock return is missing. I estimate downside beta with the same intercept-inclusive regression on $\mathcal{P}^-_{i,t}$. It requires at least 12 downside pairs and at least 36 total pairs. All input data are dated no later than t. If the history is too short, the exposure remains missing and the observation is excluded from the common regression sample. It is not assigned zero beta.

A.4 Regression Controls and Monthly Transformations

Historical mean and volatility use $\{t-11,\ldots,t\}$ and require at least six valid returns. Momentum is $\exp\{\sum_{s=t-11}^{t-1}\log(1+R_{i,s})\}-1$, also with at least six valid observations. Missing values for these historical controls use the eligible-sample median in the formation month. This rule does not apply to betas. In the empirical regressions, explanatory variables are winsorized within month at 1% and 99%. Continuous controls are then standardized. ES and contributions keep their winsorized decimal return values. These transformations apply only to the regressions; they do not change the saved neural forecasts.

B Model Implementation and Statistical Inference

Input scaling follows equation (11). The denominator includes 10^{-8} to prevent division by zero when a predictor has no variation in the training window.

B.1 Historical and Linear Benchmarks

Rolling Normal estimates each stock's mean and sample standard deviation from the 84 calendar months ending at formation. The mean requires at least one valid return and volatility at least two. If volatility is missing, I use the formation-month standard deviation across stocks. Rolling Normal standard deviations have a lower bound of 10^{-6}. For estimated mean $\widehat{\mu}$ and standard deviation $\widehat{\sigma}$, suppress the stock and formation-month indices. Write q_α for VaR and e_α for ES at tail probability α. The function Φ^{-1} is the inverse standard Normal CDF. These forecasts are

$$q_\alpha=\widehat{\mu}+\widehat{\sigma}\Phi^{-1}(\alpha),\qquad e_\alpha=\widehat{\mu}-\widehat{\sigma}\frac{\phi\{\Phi^{-1}(\alpha)\}}{\alpha}. \tag{41}$$

Linear Quantile estimates the 27 grid quantiles using firm variables, macro variables, and industry information. Each quantile is fitted with a month-weighted pinball loss. The predicted quantiles are sorted within each stock–month to prevent crossings. The reported ES forecasts and scores are calculated from this grid, with the endpoint quantiles held constant outside it. The model produces a set of quantiles, not an exact parametric density.

B.2 Fused Density Benchmarks

The fused benchmarks concatenate the 13 macroeconomic predictors m_t, the 21 firm predictors $x_{i,t}$, and a six-dimensional industry embedding $e_{i,t}$. Each model estimates

its own embedding. Define the combined input $z_{i,t}^F = (m_t', x_{i,t}', e_{i,t}')' \in \mathbb{R}^{40}$, where the superscript F labels the fused model and primes denote transpose. A single network transforms these 40 inputs through hidden layers of widths 64, 32, and 8. Each hidden layer uses layer normalization and an exponential linear unit (ELU) activation; the first two also use dropout 0.05. Thus macroeconomic and firm information enter the same learned representation before any distributional parameters are produced.

Separate linear output heads map the final eight features to component means, scale scores, and, when there is more than one component, probability scores. Softplus transforms each scale score into a positive value, to which the model adds the same 0.005 scale floor used by the structured models. Softmax transforms the probability scores into weights that sum to one. Using the superscript F to distinguish fused-model parameters, each benchmark has predictive density

$$f_{i,t}^F(r) = \sum_{k=1}^{K} \pi_{i,k,t}^F \mathcal{N}(r; \mu_{i,k,t}^F, (\sigma_{i,k,t}^F)^2). \tag{42}$$

Here $\pi_{i,k,t}^F$, $\mu_{i,k,t}^F$, and $\sigma_{i,k,t}^F$ are the fused model's component probability, mean, and standard deviation. Fused K1 sets $K = 1$ and $\pi_{i,1,t}^F = 1$, giving a conditional Normal forecast. Fused K3 sets $K = 3$ and estimates all three probabilities. Because every output head receives the combined representation, macroeconomic and firm inputs can affect every parameter. In particular, Fused K3's probabilities can differ across firms in the same month; no common monthly latent state is imposed.

Both benchmarks minimize the marginal NLL, with equal total weight for each training month:

$$\mathcal{L}_w^F = -\frac{1}{|\mathcal{T}_w|} \sum_{t \in \mathcal{T}_w} \frac{1}{N_t} \sum_{i \in \mathcal{U}_t} \log f_{i,t}^F(R_{i,t+1}). \tag{43}$$

The symbol $\mathcal{L}_w^F$ is the fused training objective in window w. It averages individual-stock negative log likelihoods within each month and then gives each training month equal weight, using the fused density. Each stock–month is weighted inversely by its month's firm count, so months with more stocks do not receive more total weight. Training uses shuffled batches of 1,024 stock observations. Validation evaluates the same month-weighted marginal NLL and selects the checkpoint with the lowest value. These benchmarks have no joint-density loss term. They use the annual rolling samples, input scaling, and optimization settings described in this section, but their batches need not contain complete monthly cross-sections.

B.3 Structured Networks and Optimization

Figure 2 summarizes the network dimensions. The macro network has hidden layers of widths 16 and 8. The firm network has hidden layers of widths 64, 32, and 8. Its inputs combine the 21 scaled firm variables and the six-dimensional industry embedding. All hidden layers use layer normalization and exponential linear unit (ELU) activations. Dropout (Srivastava et al., 2014) at rate 0.05 follows the first macro hidden layer and the first two firm hidden layers. Output layers produce component means and the values transformed by softplus into standard deviations. Here $\text{softplus}(v) = \log(1 + \exp v)$. The FiLM layer converts the eight macro features into the eight-dimensional scale change $\Delta\gamma_t$ and shift β_t defined in equation (7). Both use the 0.1 scaling in that equation, and the lower bound on component standard deviations is 0.005.

The main models use AdamW (Loshchilov and Hutter, 2019) with learning rate 0.003, weight decay 10^{-4}, gradient clipping at norm 1, and at most 500 epochs. Early stopping allows 20 epochs without a validation improvement of at least 10^{-6}. ReduceLROnPlateau multiplies the learning rate by 0.5 after five validation steps without improvement, down to a minimum of 10^{-6}. Structured training batches contain eight complete months. Months have equal total weight, no firms are subsampled, and each annual window starts with a new model. The base random seed for the main run is 2027. Each estimation window uses this base seed plus its window index w. The forecasts come from the 18 validation-selected models, without a final refit.

B.4 Forecast Quantiles and Numerical Checks

For the seven-model comparison, CRPS and lower-tail CRPS use the common quantile grid $\mathcal{G}$, a set of 27 probability levels:

$$\mathcal{G} = \{0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07, 0.08, 0.09, 0.10, 0.15, 0.25, 0.40, 0.50,$$
$$0.60, 0.75, 0.85, 0.90, 0.91, 0.92, 0.93, 0.94, 0.95, 0.96, 0.97, 0.98, 0.99\}.$$

Let Φ be the standard Normal cumulative distribution function. For a Normal distribution, quantiles use its inverse Φ^{-1}, which returns the cutoff associated with a given probability. Mixture quantiles solve

$$F_{i,t}(q) = \sum_k \pi_{i,k,t} \Phi\left(\frac{q - \mu_{i,k,t}}{\sigma_{i,k,t}}\right) = \alpha. \tag{44}$$

Here the unknown q is $q_{i,t,\alpha}$, and $F_{i,t}$ is the mixture CDF obtained by summing the weighted component CDFs. Bisection finds the cutoff by repeatedly halving an interval that brackets the desired probability. For Linear Quantile, I sort the predicted quantiles within each stock–month to prevent crossings. For CRPS, I extend the quantile grid to probabilities zero and one by holding its endpoint quantiles constant, then approximate the quantile-score integral with the trapezoidal rule. Lower-tail CRPS uses the same rule over probabilities zero to α. All seven models use the same grid, endpoint extensions, and integration rule, including models with an exact density. These grid approximations apply to CRPS and lower-tail CRPS; NLL is calculated from the predictive density, and the quantile and joint VaR–ES scores use the relevant quantile and ES forecasts.

In the main comparison, stored VaR predictions agree with independently calculated quantiles to within 5×10^{-12}.

For FZ0, the common sample satisfying $q < 0$, $e < 0$, and $e \leq q$ for all seven models retains 98.860% of stock–months at the 5% level and 97.344% at the 10% level. Table B.1 reports the share meeting these conditions for each model separately.

Table B.1: Validity of joint VaR–ES scores

Model	Valid 5% (%)	Valid 10% (%)
Rolling Normal	99.867	99.806
Linear Quantile	98.992	97.664
Fused K1	100.000	99.909
Fused K3	100.000	99.922
Structured K3	100.000	100.000
FiLM Sigma	100.000	100.000
FiLM All	100.000	100.000

Notes: A forecast is included in the FZ0 comparison when $q < 0$, $e < 0$, and $e \leq q$. Keeping only observations valid for all seven models retains 98.860% of stock–months at 5% and 97.344% at 10%.

B.5 Component Labels, ES Formulas, and Diagnostics

B.5.1 Aligning Component Labels

Mixture components can be relabeled without changing the predicted distribution (Stephens, 2000). I therefore apply the same labeling rule in each estimation window. For window w, I identify the adverse component using the fitted means in the training and validation sample:

$$A_w = \arg\min_k \frac{1}{|\mathcal{A}_w|} \sum_{s \in \mathcal{A}_w} \frac{1}{N_s} \sum_i \mu_{i,k,s}, \tag{45}$$

Here $\mathcal{A}_w$ contains the 84 months before the test year, and N_s is the number of firms in month s. I average across firms within each month and then across months. The component with the highest average mean is favorable, and the remaining component is middle. Labels remain fixed throughout the next test year. They describe fitted return distributions, rather than observed market events or identified economic shocks.

B.5.2 Computing Tail Weights and Contributions

Using $z_{i,k,t,\alpha} = (q_{i,t,\alpha} - \mu_{i,k,t})/\sigma_{i,k,t}$ and the tail weight in equation (27), the expected return in state k conditional on falling below the stock's VaR is

$$\tau_{i,k,t,\alpha} = \mu_{i,k,t} - \sigma_{i,k,t} \frac{\phi(z_{i,k,t,\alpha})}{\Phi(z_{i,k,t,\alpha})}, \tag{46}$$

where ϕ is the standard Normal density. Multiplying this conditional expected return by the tail weight gives

$$C_{i,k,t,\alpha} = \frac{\pi_{k,t}}{\alpha} \left[\mu_{i,k,t}\Phi(z_{i,k,t,\alpha}) - \sigma_{i,k,t}\phi(z_{i,k,t,\alpha})\right]. \tag{47}$$

Summing these contributions yields the exact ES identity in equation (29).

B.5.3 Component Use and Numerical Checks

For month t, define the effective component count as $K_{\text{eff},t} = 1/\sum_k \pi_{k,t}^2$. It approaches one when one component dominates and equals three for equal probabilities. Table B.2 reports its average across months, denoted $\overline{K}_{\text{eff}}$.

Table B.2: Use and numerical consistency of the aligned components

Model	$\overline{K}_{\text{eff}}$	Collapse (%)	Lowest mean (%)	ES share 5%	ES share 10%	Max ES error
Structured K3	1.870	2.31	99.92	58.70	56.30	4.4×10^{-16}
FiLM Sigma	2.159	0.00	99.57	58.42	55.19	3.3×10^{-16}
FiLM All	2.306	0.00	99.63	63.53	60.05	4.4×10^{-16}

Notes: The full-distribution diagnostics appear once. The effective component count is $K_{\text{eff},t} = 1/\sum_k \pi_{k,t}^2$. Collapse means a state probability above 0.95. Lowest mean is the average monthly fraction of firms for which the adverse component retains the lowest mean in the test sample. Each ES share is 100 times the average adverse contribution divided by average total ES at that level, giving each month equal weight. It is a share of ES, not a state probability. Max ES error is the largest absolute reconstruction error across both levels.

The fitted mixtures make substantial use of multiple components in the test sample. Their average effective component counts range from 1.87 to 2.31. A state probability

exceeds 0.95 in only five months for Structured K3 and in no months for either FiLM specification. The component labeled adverse before the test year retains the lowest mean for more than 99.5% of firms in the average test month. It contributes between 58.4% and 63.5% of average 5% ES, and between 55.2% and 60.1% of average 10% ES. These are shares of ES, not state probabilities. Contributions reconstruct total ES with an error below 10^{-12} at both levels. The decomposition thus provides a consistently defined and quantitatively important signal for the economic tests.

B.6 Cross-Stock Dependence from the Common State

All moments in this subsection are taken under the model's joint predictive distribution. The subscript t conditions on formation information $\mathcal{F}_t$. Let $S = S_{t+1}$ denote the common future state, with conditional probabilities $\pi_{k,t}$. For $i \neq j$, the law of total covariance gives

$$\begin{aligned}\mathrm{Cov}_t(R_{i,t+1}, R_{j,t+1}) = \mathbb{E}_t[\mathrm{Cov}(R_{i,t+1}, R_{j,t+1} \mid S, \mathcal{F}_t)] \\ + \mathrm{Cov}_t(\mathbb{E}[R_{i,t+1} \mid S, \mathcal{F}_t], \mathbb{E}[R_{j,t+1} \mid S, \mathcal{F}_t])\,. \end{aligned} \tag{48}$$

The first term is zero because firms' returns are independent conditional on the state and formation information. In state k, the two conditional expectations in the second term are $\mu_{i,k,t}$ and $\mu_{j,k,t}$. Their probability-weighted averages are $\widehat{m}_{i,t} = \sum_k \pi_{k,t}\mu_{i,k,t}$ and $\widehat{m}_{j,t} = \sum_k \pi_{k,t}\mu_{j,k,t}$. Expanding their covariance gives

$$\mathrm{Cov}_t(R_{i,t+1}, R_{j,t+1}) = \sum_{k=1}^{3} \pi_{k,t}(\mu_{i,k,t} - \widehat{m}_{i,t})(\mu_{j,k,t} - \widehat{m}_{j,t}), \qquad i \neq j. \tag{49}$$

Each difference in parentheses measures how a firm's mean in state k differs from its overall predictive mean. When both firms have below-average means in the same states and above-average means in the same states, the products are positive and generate positive covariance. Opposite responses can generate negative covariance. The probabilities $\pi_{k,t}$ weight these products.

The law of total variance similarly gives

$$\widehat{v}_{i,t}^2 = \underbrace{\sum_{k=1}^{3} \pi_{k,t}\sigma_{i,k,t}^2}_{\text{within-state variance}} + \underbrace{\sum_{k=1}^{3} \pi_{k,t}(\mu_{i,k,t} - \widehat{m}_{i,t})^2}_{\text{variance of state-specific means}}\,. \tag{50}$$

Correlation equals the covariance generated by differences in firms' state-specific means divided by the product of their predictive standard deviations. Component standard

deviations affect each firm's total variance and therefore correlation, even though they do not enter the cross-firm covariance directly. A common state permits nonzero covariance; its magnitude and sign depend on the conditional means and state probabilities. For example, a firm whose conditional mean is identical in all states has zero model-implied covariance with every other firm. Changing state probabilities need not change covariance or correlation monotonically.

Finally, sharing probability values is different from sharing the state realization. If each firm independently drew its own state with probabilities $\pi_{k,t}$, the joint density would instead be

$$\prod_{i \in \mathcal{U}_t} \left[\sum_{k=1}^{3} \pi_{k,t} \mathcal{N}(r_i; \mu_{i,k,t}, \sigma^2_{i,k,t}) \right]. \tag{51}$$

This expression multiplies firms' marginal densities and implies independence conditional on $\mathcal{F}_t$. In the common-state density in equation (12), the product is taken within a state before averaging over states. The fused models described in Appendix B.2 estimate individual marginal densities without specifying this common-state joint distribution; their marginal forecasts alone do not determine cross-stock correlations.

B.7 Calendar-Time Standard Errors

Forecast comparisons use monthly loss differences, portfolio tests use monthly returns, and the second-step contribution regressions use monthly estimated slopes. The monthly cross-sectional regressions follow Fama and MacBeth (1973). Petersen (2009) discusses how dependence across firms and time affects inference in financial panels. I calculate Newey–West standard errors with 12 calendar-month lags, allowing unequal variances and serial correlation across months (Newey and West, 1987).

When a comparison uses only bad months or only other months, I preserve the actual calendar gaps. For example, observations from January and April are three months apart, even if they are consecutive observations in the selected sample. Treating them as adjacent months would change the standard errors.

Each mean or regression is estimated using the selected observations only. For the standard-error calculation, the regression's regressor-weighted residuals are placed at their original calendar positions, with zero contributions from excluded months. Lagged covariance products therefore use actual calendar-month separations, with Bartlett weights up to the specified lag length. The bandwidth checks use the same convention with 3, 6, or 12 lags. Reported p-values are two-sided and use the asymptotic Normal distribution.

B.8 Paired Comparisons of PIT Diagnostics

I compare the structured models' monthly cross-sectional KS distances with those of Rolling Normal, Fused K1, and Fused K3. Each monthly difference is the structured model's KS distance minus the benchmark's, so a negative value favors the structured model. I average the differences equally across the 216 formation months and test a zero mean difference using the NW12 procedure described in Appendix B.7. Table B.3 reports all nine comparisons. These tests assess differences between models' KS diagnostics; they do not test the uniformity of the full PIT distribution or equality of calibration in every respect.

Table B.3: Paired comparisons of monthly PIT KS distances

Model	Benchmark	Δ KS	SE	p
Structured K3	Rolling Normal	0.00897	0.00643	0.163
Structured K3	Fused K1	-0.03928	0.01131	< 0.001
Structured K3	Fused K3	-0.04974	0.01258	< 0.001
FiLM Sigma	Rolling Normal	0.00378	0.00422	0.370
FiLM Sigma	Fused K1	-0.04447	0.01143	< 0.001
FiLM Sigma	Fused K3	-0.05493	0.01507	< 0.001
FiLM All	Rolling Normal	0.00581	0.00459	0.205
FiLM All	Fused K1	-0.04244	0.01086	< 0.001
FiLM All	Fused K3	-0.05290	0.01463	< 0.001

Notes: Differences are the structured model's monthly cross-sectional KS distance minus the benchmark's, averaged equally over 216 formation months. Negative values favor the structured model. Two-sided p-values test a zero mean difference using Newey–West standard errors with 12 calendar-month lags (NW12). These tests compare models' KS distances; they are separate from tests of PIT uniformity or average VaR coverage.

C Additional Regression and Sorting Results

The additional tables retain the notation from the main text. A gap is a lowest-minus-highest group difference, and pp abbreviates percentage points. Table C.3 reports the variation left after each signal is regressed on the other signal and the controls. Table C.4 uses "Beta gap" for the group difference in historical market beta and "Volatility gap" for the difference in predicted standard deviation.

For Table C.3, I regress each signal on the other signal, all controls, and industry indicators. The coefficient of determination, R^2, is the fraction of within-month signal variation explained by this linear projection; the reported $100(1 - R^2)$ is the percentage

Table C.1: Total ES and subsequent downside outcomes at both tail levels

Model	ES (%)	Outcome	All	p	Bad	p	Other	p
Structured K3	5	Crash	-0.540	< 0.001	-0.717	< 0.001	-0.495	< 0.001
Structured K3	10	Crash	-0.622	< 0.001	-0.823	< 0.001	-0.571	< 0.001
Structured K3	5	Loss beyond 10%	-0.075	< 0.001	-0.150	< 0.001	-0.056	< 0.001
Structured K3	10	Loss beyond 10%	-0.087	< 0.001	-0.173	< 0.001	-0.065	< 0.001
FiLM Sigma	5	Crash	-0.536	< 0.001	-0.685	< 0.001	-0.498	< 0.001
FiLM Sigma	10	Crash	-0.621	< 0.001	-0.792	< 0.001	-0.577	< 0.001
FiLM Sigma	5	Loss beyond 10%	-0.074	< 0.001	-0.139	< 0.001	-0.057	< 0.001
FiLM Sigma	10	Loss beyond 10%	-0.086	< 0.001	-0.161	< 0.001	-0.066	< 0.001
FiLM All	5	Crash	-0.542	< 0.001	-0.722	< 0.001	-0.496	< 0.001
FiLM All	10	Crash	-0.627	< 0.001	-0.831	< 0.001	-0.574	< 0.001
FiLM All	5	Loss beyond 10%	-0.073	< 0.001	-0.142	< 0.001	-0.055	< 0.001
FiLM All	10	Loss beyond 10%	-0.085	< 0.001	-0.165	< 0.001	-0.065	< 0.001

Notes: Monthly stock regressions include total ES, historical size and return controls, and industry indicators; adverse contribution is omitted. Entries are average monthly ES coefficients for all, bad, or other months. A one-percentage-point increase in ES corresponds to the reported change in the outcome in percentage points. Both ES levels use the same sample with complete historical controls. Two-sided p-values use Newey–West standard errors with 12 calendar-month lags.

Table C.2: Contribution slopes for returns and losses beyond the 10% threshold at both ES levels

Model	Outcome	δ_C: 5%	SE	p	δ_C: 10%	SE	p
Structured K3	Raw return	0.853	0.255	< 0.001	1.272	0.400	0.001
Structured K3	Loss beyond 10%	-0.317	0.142	0.026	-0.489	0.212	0.021
FiLM Sigma	Raw return	0.455	0.099	< 0.001	0.731	0.160	< 0.001
FiLM Sigma	Loss beyond 10%	-0.165	0.051	0.001	-0.274	0.086	0.002
FiLM All	Raw return	0.471	0.111	< 0.001	0.679	0.173	< 0.001
FiLM All	Loss beyond 10%	-0.194	0.068	0.004	-0.276	0.100	0.006

Notes: The regressions use the full control set in Table 9. Each entry is the difference in monthly contribution coefficients between bad market months and other months, adjusted for annual estimation-window indicators. Loss beyond 10% is $\max\{-R - 0.10, 0\}$. Units are outcome percentage points per one-percentage-point increase in contribution. Positive return coefficients and negative coefficients for losses beyond 10% indicate worse relative outcomes for stocks with more negative contributions in bad market months. Two-sided p-values use Newey–West standard errors with 12 calendar-month lags.

Table C.3: ES and adverse-contribution variance left after controlling for other risk measures

Model	Signal	Variance retained (%): 5% ES	Variance retained (%): 10% ES
Structured K3	ES	0.0259	0.0201
Structured K3	Contribution	7.5217	5.8007
FiLM Sigma	ES	0.0365	0.0264
FiLM Sigma	Contribution	9.4194	6.6043
FiLM All	ES	0.0340	0.0259
FiLM All	Contribution	8.8752	6.5739

Notes: Within each month and ES level, I regress each signal on the other signal, all controls, and industry indicators. Entries are $100(1 - R^2)$, averaged across months with equal weights. They report the percentage of the winsorized signal variance left after the linear adjustment.

Table C.4: Characteristics of groups sorted by contribution residuals at both ES levels

Model	ES (%)	Market group	ES gap (pp)	Beta gap	Volatility gap (pp)
Structured K3	5	Bad months	-0.149	-0.034	0.064
Structured K3	5	Other months	0.041	-0.025	-0.031
Structured K3	10	Bad months	-0.197	-0.026	0.089
Structured K3	10	Other months	0.103	-0.027	-0.073
FiLM Sigma	5	Bad months	-0.252	-0.023	0.090
FiLM Sigma	5	Other months	0.019	-0.026	-0.037
FiLM Sigma	10	Bad months	-0.350	-0.015	0.154
FiLM Sigma	10	Other months	-0.103	-0.019	0.024
FiLM All	5	Bad months	0.364	-0.049	-0.213
FiLM All	5	Other months	-0.052	-0.024	-0.003
FiLM All	10	Bad months	0.007	-0.040	-0.051
FiLM All	10	Other months	-0.145	-0.020	0.055

Notes: Stocks are ranked by their contribution residuals. Entries are the bottom decile's average characteristic minus the top decile's, averaged across months with equal weights. ES and predicted-volatility gaps are in percentage points; market beta remains in its original units. The monthly linear adjustment removes the fitted relation between contribution and the controls. The entries describe the characteristics of the resulting stock groups.

left in its residual. At 5% ES, the average share of variation left in the residual is about 7.5%–9.4% for contribution and 0.026%–0.037% for total ES. At 10% ES, the corresponding ranges are 5.8%–6.6% and 0.020%–0.026%. Total ES is therefore very closely related to the other regressors. The residual sorts complement the coefficient estimates by comparing stock groups formed from the remaining contribution variation. Table C.4 reports the groups' average characteristics.

D Market Definitions and Bandwidth Robustness

Table D.1: Alternative bad-market rules at both ES levels

Model	Market excess rule	Bad months	δ_C: 5%	p	δ_C: 10%	p
Structured K3	Bottom 20%	44	-4.114	< 0.001	-6.239	< 0.001
Structured K3	$\leq -2.5\%$	44	-4.114	< 0.001	-6.239	< 0.001
Structured K3	$\leq -3\%$	40	-4.181	< 0.001	-6.439	< 0.001
Structured K3	$\leq -5\%$	25	-2.706	0.002	-4.408	< 0.001
FiLM Sigma	Bottom 20%	44	-2.386	< 0.001	-3.767	< 0.001
FiLM Sigma	$\leq -2.5\%$	44	-2.386	< 0.001	-3.767	< 0.001
FiLM Sigma	$\leq -3\%$	40	-2.313	< 0.001	-3.689	< 0.001
FiLM Sigma	$\leq -5\%$	25	-1.797	< 0.001	-3.011	< 0.001
FiLM All	Bottom 20%	44	-2.343	< 0.001	-3.495	< 0.001
FiLM All	$\leq -2.5\%$	44	-2.343	< 0.001	-3.495	< 0.001
FiLM All	$\leq -3\%$	40	-2.287	< 0.001	-3.449	< 0.001
FiLM All	$\leq -5\%$	25	-1.873	< 0.001	-2.970	< 0.001

Notes: All specifications use full controls, signed adverse contribution at the indicated ES level, capped at its monthly 1st and 99th percentiles, and estimation-window effects. The individual stock crash cutoff remains -10%. Both levels use the same market dates under each rule. Months receive equal weight. Two-sided p-values use NW12 standard errors.

The alternative market rules use market excess returns. The individual stock crash cutoff stays at a raw return below -10%. Using the 10% tail changes ES and its decomposition from the 5% version, but does not change this crash cutoff. These checks use the same regression sample with all controls.

Table D.2: Sensitivity to the Newey–West lag length at both ES levels

Model	Lags	δ_C: 5%	SE	p	δ_C: 10%	SE	p
Structured K3	3	-4.114	1.003	< 0.001	-6.239	1.556	< 0.001
Structured K3	6	-4.114	1.131	< 0.001	-6.239	1.745	< 0.001
Structured K3	12	-4.114	1.227	< 0.001	-6.239	1.883	< 0.001
FiLM Sigma	3	-2.386	0.362	< 0.001	-3.767	0.548	< 0.001
FiLM Sigma	6	-2.386	0.349	< 0.001	-3.767	0.529	< 0.001
FiLM Sigma	12	-2.386	0.345	< 0.001	-3.767	0.544	< 0.001
FiLM All	3	-2.343	0.335	< 0.001	-3.495	0.520	< 0.001
FiLM All	6	-2.343	0.347	< 0.001	-3.495	0.544	< 0.001
FiLM All	12	-2.343	0.363	< 0.001	-3.495	0.580	< 0.001

Notes: Point estimates use the same full-control monthly coefficients and the same 44 bad months at each tail level. Only the number of calendar-month Newey–West lags changes. Reported p-values are two-sided.

E Random Seeds and Training Protocols

E.1 Training Dynamics and Checkpoint Selection

The saved logs allow a comparison of the five neural models' validation dynamics across all 18 rolling windows. The fused models minimize marginal NLL. The structured specifications minimize the combination of normalized joint and marginal NLL in equation (15). To compare a common quantity, Figure E.1 uses validation marginal NLL for all five neural models, including the marginal term recorded separately for the structured specifications. Each curve shows the median change from the end of epoch 1 across the 18 windows. All models have observations through epoch 20, so no curve is extended or filled after early stopping.

The structured specifications show continuing median validation improvement over this interval. Their validation losses also have smaller typical changes between consecutive epochs. For window w, define

$$V_w = \text{median}_{n=7,\ldots,20} \left| \mathcal{L}_{w,n}^{M,\text{val}} - \mathcal{L}_{w,n-1}^{M,\text{val}} \right|, \tag{52}$$

Here n indexes training epochs. The quantity $\mathcal{L}_{w,n}^{M,\text{val}}$ is the marginal NLL on the validation months of window w after epoch n; M means marginal and val means validation. V_w is the median of the 14 changes between successive epochs from epoch 6 through epoch 20. The loss plot also retains the earlier epochs. A smaller V_w means smaller typical epoch-to-epoch changes in validation marginal NLL. In the main run, the median V_w is 0.02226 and 0.02357 for Fused K1 and Fused K3, compared with 0.00841, 0.00593,

and 0.00785 for Structured K3, FiLM Sigma, and FiLM All. Table E.1 shows the same ordering between each structured specification and both fused benchmarks in each of the two additional validation seeds.

Table E.1: Validation dynamics and checkpoint selection across random seeds

Seed	Model	Epoch-1 checkpoints	Median checkpoint	Median V_w
2027	Fused K1	10/18	1.0	0.02226
2027	Fused K3	11/18	1.0	0.02357
2027	Structured K3	0/18	30.0	0.00841
2027	FiLM Sigma	0/18	26.0	0.00593
2027	FiLM All	0/18	21.5	0.00785
2026	Fused K1	14/18	1.0	0.03192
2026	Fused K3	7/18	3.0	0.02600
2026	Structured K3	0/18	22.0	0.00908
2026	FiLM Sigma	0/18	21.0	0.00923
2026	FiLM All	0/18	23.5	0.01044
42	Fused K1	7/18	4.0	0.02988
42	Fused K3	10/18	1.0	0.02409
42	Structured K3	0/18	20.0	0.00943
42	FiLM Sigma	0/18	21.5	0.00938
42	FiLM All	0/18	22.5	0.00925

Notes: Each row summarizes all 18 rolling estimation windows. The checkpoint is selected by marginal validation NLL for Fused and combined validation loss for the structured specifications. Epoch 1 follows one complete training epoch. For each window, $V_w = \text{median}_{n=7,\ldots,20} |\mathcal{L}_{w,n}^{M,\text{val}} - \mathcal{L}_{w,n-1}^{M,\text{val}}|$, where $\mathcal{L}^{M,\text{val}}$ is validation marginal NLL for each of the five neural models; the last column is the median across windows. The statistic summarizes the magnitude of 14 adjacent changes. The fixed-40-epoch run has no separate validation trace. Objectives, batches, and update counts per epoch differ across the fused and structured procedures.

Checkpoint selection gives a related view. Fused K1 and Fused K3 retain epoch 1 in 10 and 11 windows in the main run. None of the structured specifications retains epoch 1 in any of the three validation seeds. The median retained epoch is 21.5–30 for the three structured specifications in the main run, compared with 1 for both fused models. Checkpoint selection uses each model's validation criterion.

Figure E.2 examines the two terms of the structured objective within each window. Both terms depend on the component probabilities, means, and scales; the joint term is not a separate loss for the probability network. The figure compares changes in the two losses between consecutive epochs in both training and validation. In the main run, the joint term varies less in 18, 16, and 13 training windows for Structured K3, FiLM Sigma, and FiLM All, respectively. The corresponding validation counts are 14, 13, and

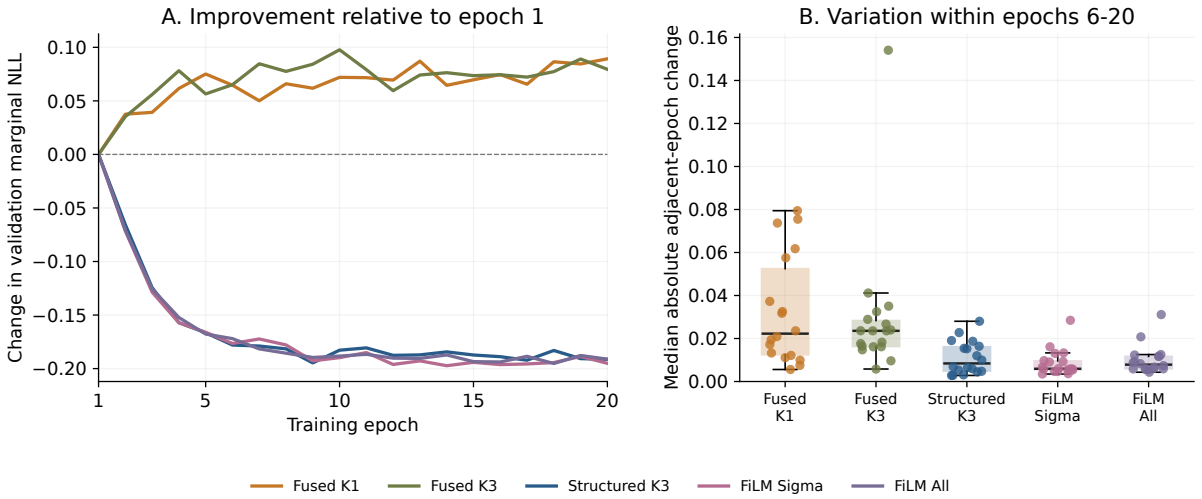


Figure E.1: Validation dynamics across the 18 rolling windows

Notes: The main run uses seed 2027. Panel A shows the across-window median of each window's change in validation marginal NLL relative to epoch 1, without smoothing. Negative changes indicate improvement. Panel B shows each window's V_w from equation (52); boxes summarize the distribution across windows and are not confidence intervals. The first epoch is a completed training epoch. All windows have at least 20 recorded epochs.

10. Across all three validation seeds, the training comparison favors the joint term in at least 13 of 18 windows for each specification. The joint term shows smaller changes more often in training than in validation. Training losses are recorded while dropout is active and model parameters change during the epoch. Validation losses are evaluated after the epoch.

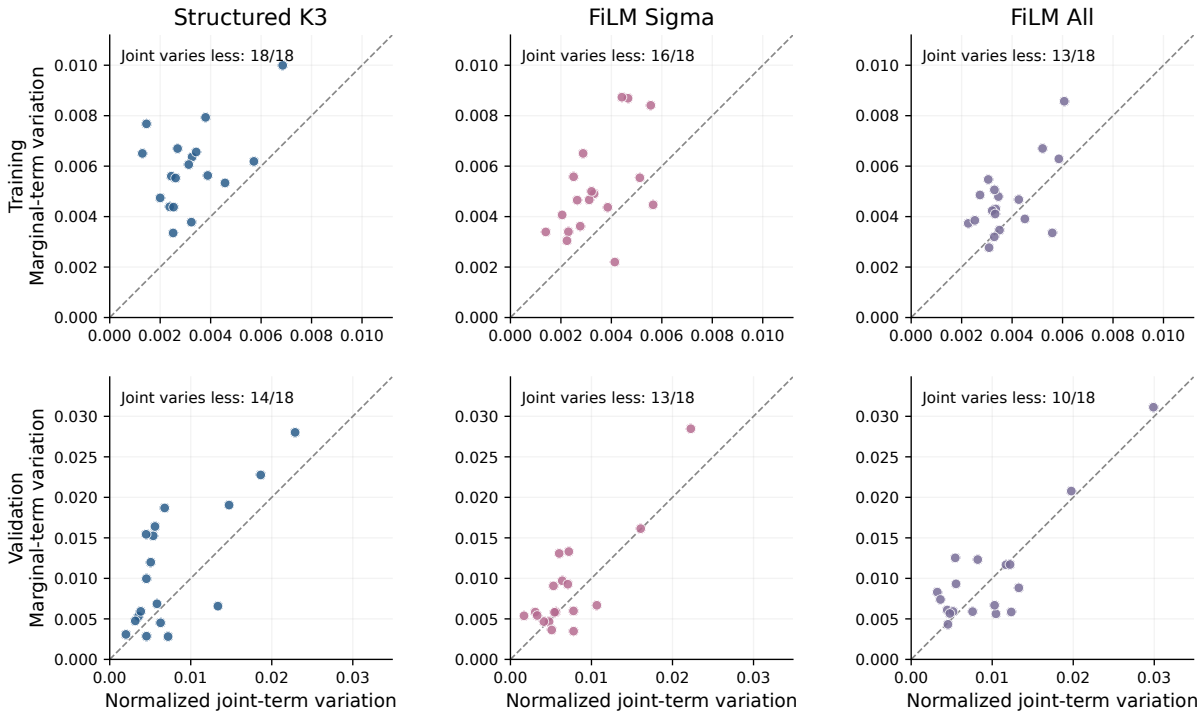


Figure E.2: Paired variation of the joint and marginal loss terms

Notes: Each point represents one of the 18 windows in the main run. Both coordinates use the median absolute adjacent-epoch change over epochs 6–20. The joint term is already normalized by the average training cross-section size. Above the dashed 45-degree line, the joint term varies less than the marginal term; below it, the reverse holds. Columns show the three structured specifications; rows show training and validation. Coordinate scales are common across models within a row.

These diagnostics show more sustained validation improvement and smaller typical changes under the structured training procedures. The fused procedure uses batches of 1,024 stock observations, while the structured procedures use batches of eight complete monthly cross-sections and a different loss. An epoch therefore also contains different numbers of parameter updates. These patterns complement the forecast evidence by showing how the structured models learn across rolling windows.

E.2 Random-Seed Sensitivity

Tables E.2 and E.3 report contribution slopes and residual gaps at 5% and 10% ES for all four runs, including the fixed-epoch check discussed below. Seeds 2027, 2026, and 42 use the same validation procedure, inputs, and model settings. The main text presents base seed 2027. Results for all three validation runs are reported to assess sensitivity to initialization. Each annual window adds its index w to the reported base seed. Across the three validation runs and both ES levels, every bad-market residual gap is positive and significant at 1%. For 5% ES, all other-month gaps have $p > 0.05$. For 10% ES, the FiLM Sigma other-month gap with seed 2026 is negative with $p = 0.039$; the remaining eight have $p > 0.05$. The coefficient magnitudes vary across runs. The consistent finding is their sign and the larger crash-rate gaps in bad months. Each run evaluates the same stock outcomes, with results reported separately.

Table E.5 compares each structured model with the fused benchmarks within each run. All three structured specifications have lower NLL, CRPS, and FZ0 at both the 5% and 10% levels than both benchmarks under every training configuration. The leading structured specification can change with the score and initialization. The common improvement across variants is the main result.

Table E.6 compares the range across the three structured–structured pairs with the range across the six structured–fused pairs. For each run, each ES tail level, and each extreme stock decile, the lowest within-family average overlap exceeds the highest structured–fused average overlap. The overlaps show that the structured variants select similar stocks on the same monthly universe.

E.3 Fixed-Epoch Training

The fixed-epoch results appear alongside the validation runs in Tables E.2, E.3, and E.5. The fixed-40-epoch run uses seed 2027 and combines training and validation into 84 months. It uses a constant learning rate, no validation checkpoint selection, and the final model at epoch 40. The architecture, predictor set, and test period are unchanged. This provides a

Table E.2: Contribution and residual sorts across all four runs: 5% ES

Seed / protocol	Model	δ_C	p	Bad gap	p	Other gap	p
2027 / Val.	Structured K3	-4.114	< 0.001	6.449	< 0.001	0.091	0.836
2027 / Val.	FiLM Sigma	-2.386	< 0.001	7.141	< 0.001	-0.406	0.294
2027 / Val.	FiLM All	-2.343	< 0.001	6.588	< 0.001	0.093	0.806
2026 / Val.	Structured K3	-3.208	< 0.001	4.876	< 0.001	-0.572	0.156
2026 / Val.	FiLM Sigma	-1.931	< 0.001	4.553	< 0.001	-0.718	0.061
2026 / Val.	FiLM All	-1.483	< 0.001	3.581	< 0.001	-0.087	0.829
42 / Val.	Structured K3	-1.928	< 0.001	3.818	< 0.001	0.197	0.587
42 / Val.	FiLM Sigma	-1.589	< 0.001	4.208	< 0.001	0.066	0.890
42 / Val.	FiLM All	-1.730	< 0.001	4.559	< 0.001	0.201	0.618
2027 / 40	Structured K3	-4.620	< 0.001	5.678	< 0.001	0.489	0.290
2027 / 40	FiLM Sigma	-3.249	< 0.001	6.547	< 0.001	0.351	0.262
2027 / 40	FiLM All	-2.218	< 0.001	7.033	< 0.001	0.710	0.038

Notes: All regressions include all controls and use the indicated ES level. Gaps are bottom-minus-top decile differences in crash rates, with stocks ranked by their contribution residuals. Gap units are percentage points. The slope difference δ_C is measured in crash-probability percentage points per one-percentage-point increase in signed contribution. Val. denotes validation-based checkpoint selection; 40 denotes fixed-40-epoch training. Each run evaluates the same stock outcomes separately. Months receive equal weight. Two-sided p-values use NW12 standard errors. The reported seed is the base seed; each annual window adds its index w.

Table E.3: Contribution and residual sorts across all four runs: 10% ES

Seed / protocol	Model	δ_C	p	Bad gap	p	Other gap	p
2027 / Val.	Structured K3	-6.239	< 0.001	6.463	< 0.001	-0.003	0.994
2027 / Val.	FiLM Sigma	-3.767	< 0.001	7.596	< 0.001	-0.205	0.624
2027 / Val.	FiLM All	-3.495	< 0.001	6.207	< 0.001	0.162	0.656
2026 / Val.	Structured K3	-5.112	< 0.001	5.146	< 0.001	-0.603	0.138
2026 / Val.	FiLM Sigma	-2.795	< 0.001	4.047	< 0.001	-0.835	0.039
2026 / Val.	FiLM All	-2.391	< 0.001	3.735	< 0.001	0.048	0.910
42 / Val.	Structured K3	-3.290	< 0.001	4.346	< 0.001	0.179	0.608
42 / Val.	FiLM Sigma	-2.314	< 0.001	3.946	0.001	-0.010	0.979
42 / Val.	FiLM All	-2.664	< 0.001	4.867	< 0.001	0.161	0.676
2027 / 40	Structured K3	-7.932	< 0.001	6.406	< 0.001	0.244	0.619
2027 / 40	FiLM Sigma	-4.971	< 0.001	6.659	< 0.001	0.057	0.847
2027 / 40	FiLM All	-3.476	< 0.001	6.888	< 0.001	0.463	0.194

Notes: All regressions include all controls and use the indicated ES level. Gaps are bottom-minus-top decile differences in crash rates, with stocks ranked by their contribution residuals. Gap units are percentage points. The slope difference δ_C is measured in crash-probability percentage points per one-percentage-point increase in signed contribution. Val. denotes validation-based checkpoint selection; 40 denotes fixed-40-epoch training. Each run evaluates the same stock outcomes separately. Months receive equal weight. Two-sided p-values use NW12 standard errors. The reported seed is the base seed; each annual window adds its index w.

Table E.4: Forecast scores across validation seeds at both tail levels

Seed	Model	CRPS	Tail CRPS 5%	Tail CRPS 10%	FZ0 5%	FZ0 10%
2027	Structured K3	0.072225	0.015872	0.026502	-1.364756	-1.563254
2027	FiLM Sigma	0.071225	0.015769	0.026332	-1.370090	-1.565342
2027	FiLM All	0.071661	0.015923	0.026527	-1.351581	-1.548862
2026	Structured K3	0.071721	0.016088	0.026698	-1.340979	-1.548172
2026	FiLM Sigma	0.071455	0.015922	0.026558	-1.360099	-1.556523
2026	FiLM All	0.071079	0.015943	0.026535	-1.355274	-1.555597
42	Structured K3	0.071052	0.015770	0.026256	-1.365504	-1.560350
42	FiLM Sigma	0.070675	0.015794	0.026289	-1.362932	-1.558994
42	FiLM All	0.071153	0.015981	0.026569	-1.349467	-1.543713

Notes: Scores are equal-month averages on each run's common seven-model sample. The main base seed is 2027. CRPS evaluates the whole distribution; tail CRPS and FZ0 use the indicated tail probability. Lower scores are preferred. The reported seed is the base seed; each annual window adds its index w.

Table E.5: Neural forecast comparisons across all four runs and both tail levels

Seed / protocol	Model	NLL	CRPS	FZ0 5%	FZ0 10%
2027 / Val.	Fused K1	-0.50290	0.07803	-1.21732	-1.40765
2027 / Val.	Fused K3	-0.50514	0.08132	-1.17743	-1.36060
2027 / Val.	Structured K3	-0.64062	0.07222	-1.36476	-1.56325
2027 / Val.	FiLM Sigma	-0.66082	0.07123	-1.37009	-1.56534
2027 / Val.	FiLM All	-0.64746	0.07166	-1.35158	-1.54886
2026 / Val.	Fused K1	-0.44632	0.08123	-1.17600	-1.40285
2026 / Val.	Fused K3	-0.49974	0.08110	-1.03109	-1.32794
2026 / Val.	Structured K3	-0.64470	0.07172	-1.34098	-1.54817
2026 / Val.	FiLM Sigma	-0.65373	0.07146	-1.36010	-1.55652
2026 / Val.	FiLM All	-0.65848	0.07108	-1.35527	-1.55560
42 / Val.	Fused K1	-0.43570	0.08135	-1.05452	-1.31122
42 / Val.	Fused K3	-0.47375	0.08232	-1.04166	-1.30940
42 / Val.	Structured K3	-0.66042	0.07105	-1.36550	-1.56035
42 / Val.	FiLM Sigma	-0.67047	0.07068	-1.36293	-1.55899
42 / Val.	FiLM All	-0.65947	0.07115	-1.34947	-1.54371
2027 / 40	Fused K1	-0.40555	0.07778	-0.96724	-1.29369
2027 / 40	Fused K3	-0.50349	0.07956	-0.91684	-1.22740
2027 / 40	Structured K3	-0.62356	0.07350	-1.31337	-1.52002
2027 / 40	FiLM Sigma	-0.63970	0.07242	-1.32419	-1.53758
2027 / 40	FiLM All	-0.65375	0.07174	-1.34871	-1.55513

Notes: Each run uses its common seven-model forecast sample and its tail-level-specific common FZ0 validity masks. Lower losses are preferred. Val. denotes validation selection; 40 denotes fixed-40 training. Runs are reported separately. The reported seed is the base seed; each annual window adds its index w.

Table E.6: Overlap in extreme-ES stock selections across runs

Seed / protocol	ES tail (%)	Low: within	Low: vs. fused	High: within	High: vs. fused
2027 / Val.	5	88.2–90.7	67.7–77.6	91.2–93.7	72.5–74.7
2027 / Val.	10	88.0–90.6	66.6–77.2	90.8–93.4	69.7–73.2
2026 / Val.	5	90.2–90.8	71.0–75.7	91.4–93.0	71.3–73.2
2026 / Val.	10	90.1–90.8	68.6–74.8	91.2–92.8	68.9–71.6
42 / Val.	5	88.6–91.1	70.4–77.8	90.7–92.3	69.1–72.5
42 / Val.	10	88.5–91.1	68.9–77.2	90.5–92.1	67.3–70.7
2027 / 40	5	90.8–92.0	72.5–76.9	92.3–94.0	69.8–72.4
2027 / 40	10	90.7–91.8	71.0–76.2	91.8–93.7	66.9–69.9

Notes: Entries give the range of average overlap percentages across model pairs, using the measure in equation (23). Within covers the three pairs among Structured K3, FiLM Sigma, and FiLM All; vs. fused covers their six comparisons with Fused K1 and Fused K3. Low and High refer to the lowest and highest ES stock deciles, each approximately 10% of the monthly cross-section, regardless of the ES tail level. Each run uses the common seven-model forecast universe of 480,446 stock–months over 216 months, before the additional portfolio eligibility restrictions. Val. denotes validation selection; 40 denotes fixed-40-epoch training. The table summarizes similarity in selected stock membership.

check under an alternative training protocol.

The contribution slope differences with all controls remain negative at both ES levels, with all six p-values below 0.001. The window-adjusted residual bad-minus-other crash gaps are 5.01, 6.40, and 6.89 percentage points at 5% ES, and 5.91, 6.86, and 6.97 points at 10% ES, also with $p < 0.001$. These results preserve the main pattern of greater vulnerability in bad market months even though forecast scores and coefficient magnitudes change.

E.4 Mixture Forecasts and Portfolio Results Across Runs

Tables E.7 and E.8 repeat the mixture–Normal comparison over the full 216-month period for each run. All three mixtures have significantly lower NLL than Normal distributions with the same mean and variance in every run. The CRPS and FZ0 results vary across runs. Each comparison uses the same 480,446 stock–months as Section 3.4. Tables E.9 and E.10 show positive bad-market ES spreads under both weighting schemes in all four runs. The repeated bad-market differences support the usefulness of total-ES rankings.

Table E.7: Mixture and Normal density forecasts across runs

Seed / protocol	Model	NLL difference	p	CRPS difference	p
2027 / Val.	Structured K3	-0.012954	< 0.001	0.000002	0.915
2027 / Val.	FiLM Sigma	-0.011006	< 0.001	-0.000020	0.378
2027 / Val.	FiLM All	-0.011405	0.007	-0.000001	0.975
2026 / Val.	Structured K3	-0.018686	0.003	-0.000006	0.658
2026 / Val.	FiLM Sigma	-0.013610	< 0.001	-0.000023	0.165
2026 / Val.	FiLM All	-0.014484	< 0.001	-0.000032	0.035
42 / Val.	Structured K3	-0.014114	< 0.001	-0.000027	0.193
42 / Val.	FiLM Sigma	-0.011668	< 0.001	-0.000041	< 0.001
42 / Val.	FiLM All	-0.007113	< 0.001	-0.000014	0.484
2027 / 40	Structured K3	-0.017707	< 0.001	-0.000036	0.226
2027 / 40	FiLM Sigma	-0.014806	0.001	-0.000030	0.125
2027 / 40	FiLM All	-0.013602	0.002	-0.000013	0.535

Notes: Each run uses the same 480,446 stock–months over the full 216-month evaluation period. Each Normal forecast has the same mean and variance as its associated mixture. Differences are mixture loss minus Normal loss; negative values favor the mixture. Val. denotes validation selection; 40 denotes fixed-40-epoch training. Two-sided p-values use Newey–West standard errors with 12 calendar-month lags.

Table E.8: Mixture and Normal VaR–ES forecasts across runs

Seed / protocol	Model	FZ0 5% difference	p	FZ0 10% difference	p
2027 / Val.	Structured K3	-0.000460	0.756	-0.000127	0.863
2027 / Val.	FiLM Sigma	-0.000769	0.545	-0.000467	0.512
2027 / Val.	FiLM All	-0.002680	0.276	-0.001445	0.311
2026 / Val.	Structured K3	-0.003041	0.193	0.000167	0.897
2026 / Val.	FiLM Sigma	-0.000313	0.745	0.001196	0.263
2026 / Val.	FiLM All	-0.001457	0.477	-0.000083	0.941
42 / Val.	Structured K3	-0.000246	0.919	0.000101	0.934
42 / Val.	FiLM Sigma	-0.000957	0.479	-0.000315	0.659
42 / Val.	FiLM All	-0.002727	0.207	-0.001076	0.328
2027 / 40	Structured K3	-0.006202	0.041	-0.000701	0.679
2027 / 40	FiLM Sigma	-0.005157	0.022	-0.001807	0.028
2027 / 40	FiLM All	-0.001444	0.330	0.000075	0.954

Notes: Each run uses the same 480,446 stock–months over the full 216-month evaluation period. Each Normal forecast has the same mean and variance as its associated mixture. Differences are mixture loss minus Normal loss; negative values favor the mixture. Val. denotes validation selection; 40 denotes fixed-40-epoch training. Two-sided p-values use Newey–West standard errors with 12 calendar-month lags.

Table E.9: ES portfolio results across all four runs: 5% ES

Seed / protocol	Model	Bad EW	Bad VW
2027 / Val.	Structured K3	6.184	5.887
2027 / Val.	FiLM Sigma	6.080	5.446
2027 / Val.	FiLM All	5.959	5.210
2026 / Val.	Structured K3	5.953	5.526
2026 / Val.	FiLM Sigma	6.048	5.651
2026 / Val.	FiLM All	5.735	5.168
42 / Val.	Structured K3	6.136	5.728
42 / Val.	FiLM Sigma	6.127	5.774
42 / Val.	FiLM All	6.058	5.349
2027 / 40	Structured K3	6.498	6.297
2027 / 40	FiLM Sigma	6.455	6.144
2027 / 40	FiLM All	6.474	6.154

Notes: Entries are monthly percentages for the highest-minus-lowest ES stock-decile portfolio. Bad EW/VW are mean returns in the same 44 bad months. The ES level is fixed within each table. Months receive equal weight.

Table E.10: ES portfolio results across all four runs: 10% ES

Seed / protocol	Model	Bad EW	Bad VW
2027 / Val.	Structured K3	6.230	6.109
2027 / Val.	FiLM Sigma	6.097	5.516
2027 / Val.	FiLM All	5.905	5.112
2026 / Val.	Structured K3	5.970	5.446
2026 / Val.	FiLM Sigma	6.071	5.741
2026 / Val.	FiLM All	5.756	5.193
42 / Val.	Structured K3	6.125	5.678
42 / Val.	FiLM Sigma	6.142	5.720
42 / Val.	FiLM All	6.082	5.388
2027 / 40	Structured K3	6.428	6.277
2027 / 40	FiLM Sigma	6.397	6.013
2027 / 40	FiLM All	6.487	6.288

Notes: Entries are monthly percentages for the highest-minus-lowest ES stock-decile portfolio. Bad EW/VW are mean returns in the same 44 bad months. The ES level is fixed within each table. Months receive equal weight.